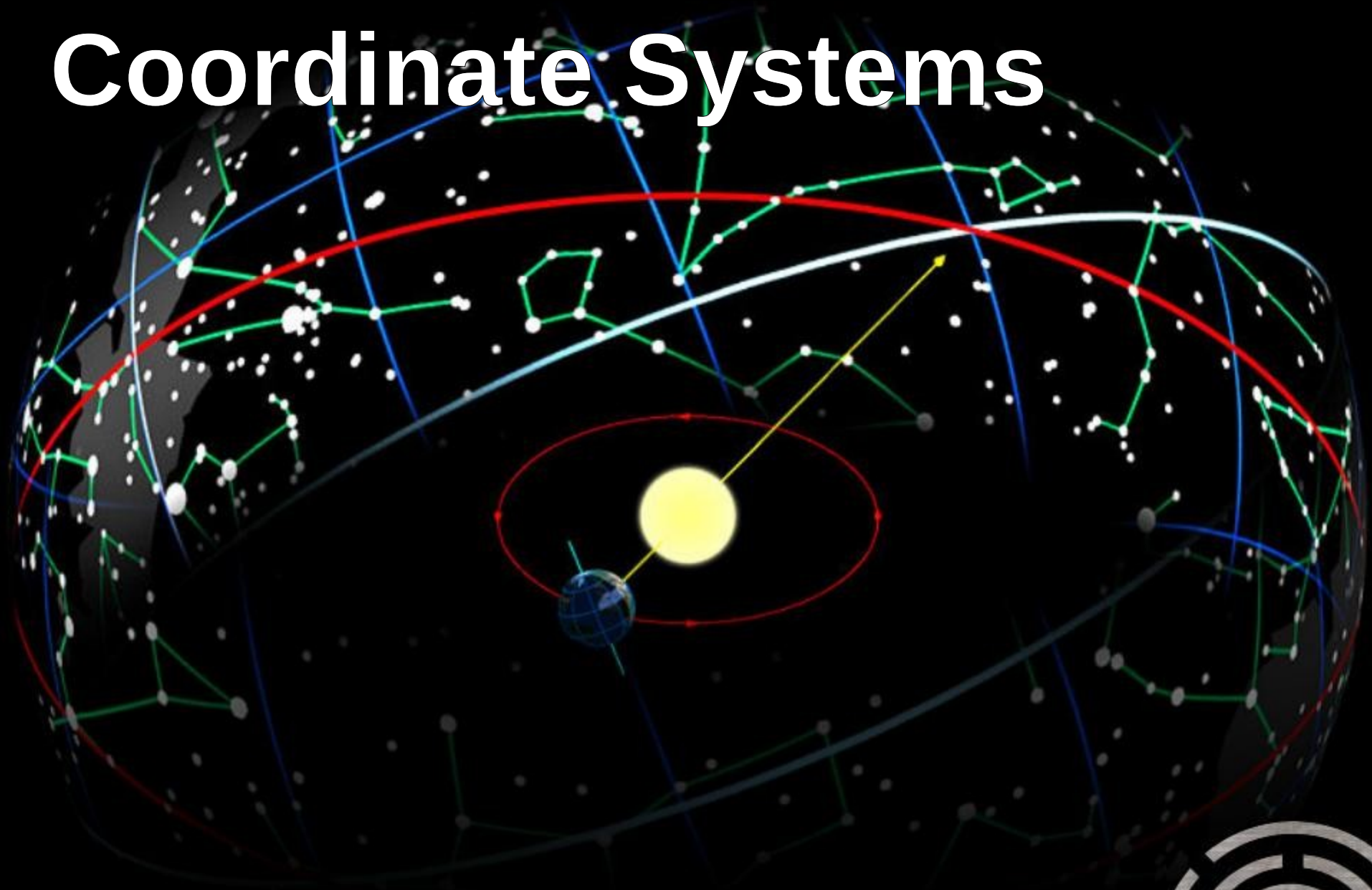
Julius-Maximillians-
UNIVERSITÄT
WÜRZBURG

Aerospace Information Technology

Coordinate Systems



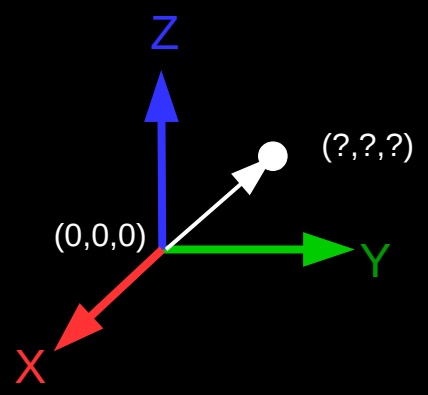
Pose: Position & Attitude: 3D + 3D = 6 DoF

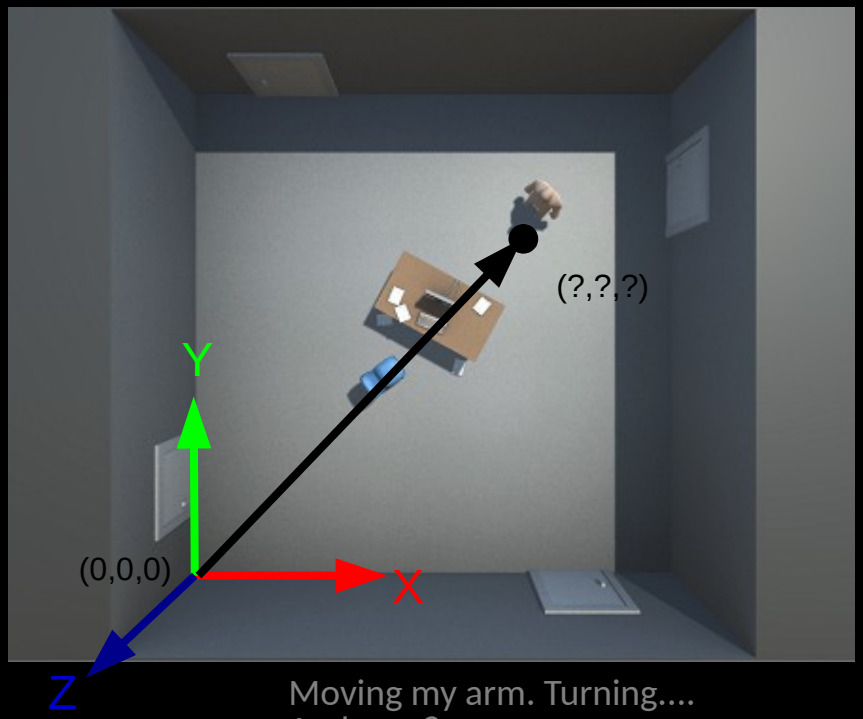
DoF = Degree of Freedom → „Pose“

Δ Position	=	Translation
Δ Attitude	=	Rotation
Δ Pose	=	Translation + Rotation

d(position)/dt	=	Speed
d(attitude)/dt	=	Angular Velocity
d(pose)/dt	=	Speed + Angular Velocity

Where am I?





My Position?

Here, for sure!
And for 90 minutes: +/- 8 m²

My Attitude?

I'm standing, for sure!
And this indepent from my position!

Moving my arm. Turning....
And now?
Where am I / a point in me?
Which one?
Nose? Finger?
Belly button?
Center of mass (moving arm)?

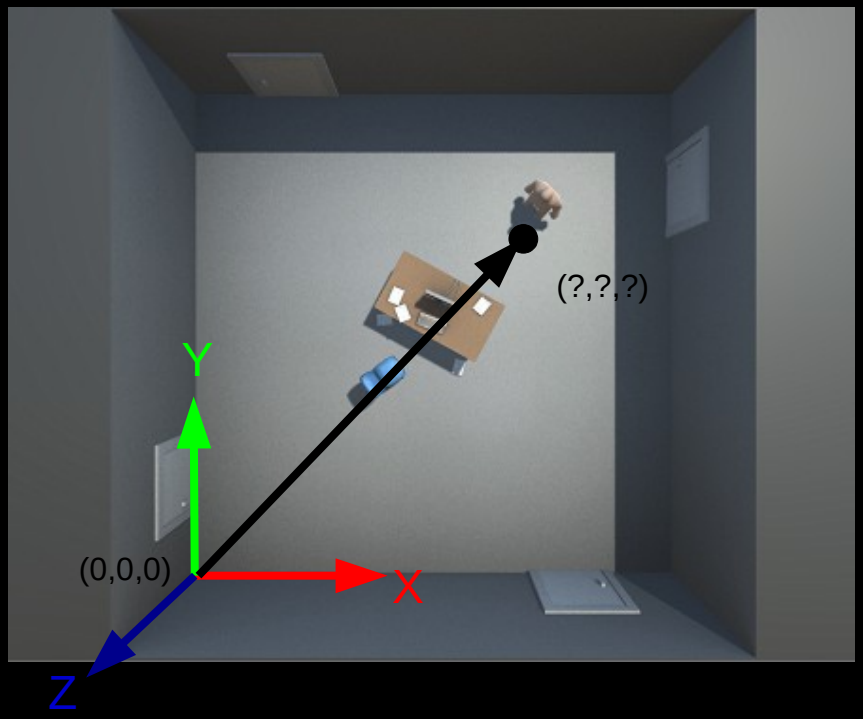
„Everything is Relative“

Where is my laptop?

The button „A“ relative to the laptop,
I can move the laptop in the room,
Room gets turned with the earth
Earth spins around the sun
Sun spins around...

Is it good like that? Why?

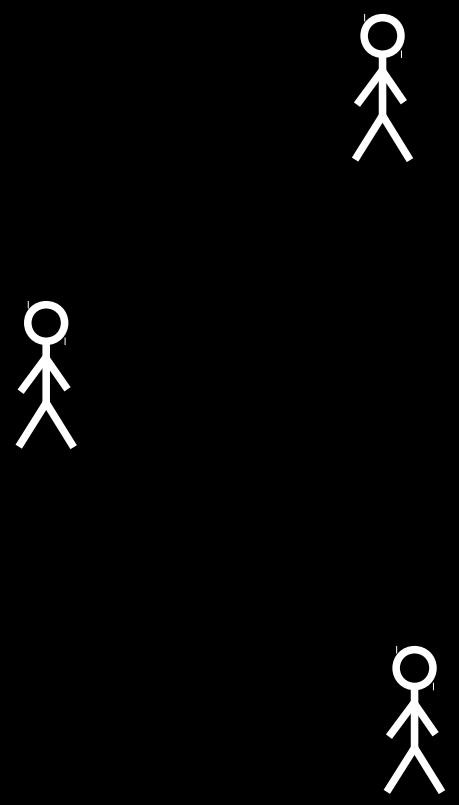
Do we want a coordinate system independent from everything around it?
=> Absolute ...xxx... as a reference?



Attitude: Angle Between What?

My Attitude?

I'm standing, for sure!
And this indepent from my position!



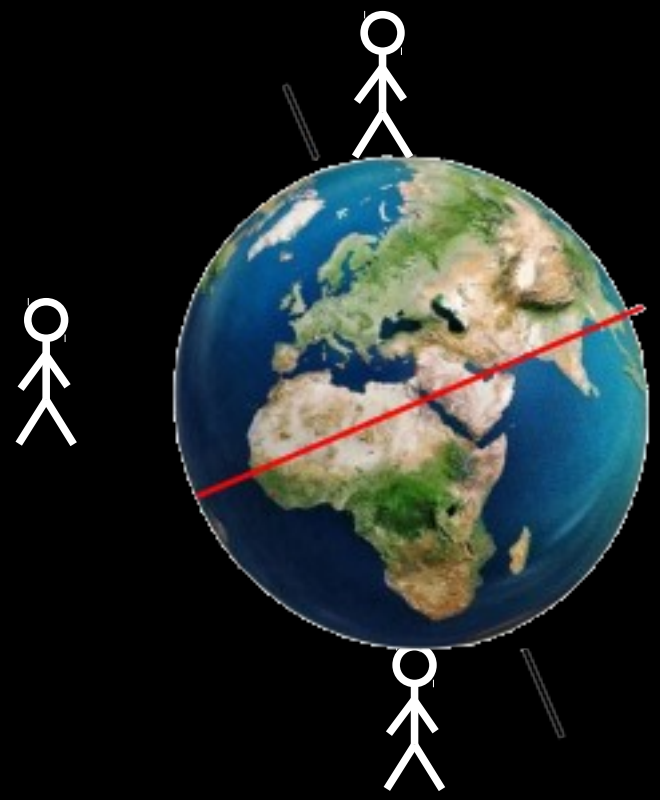
Does those 3 have the same attitude?

Attitude: Angle Between What?

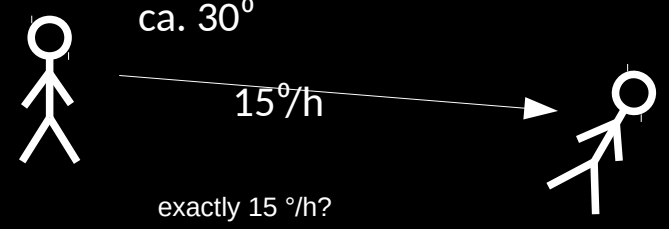
My Attitude?

I'm standing, for sure!

And this indepent from my position!

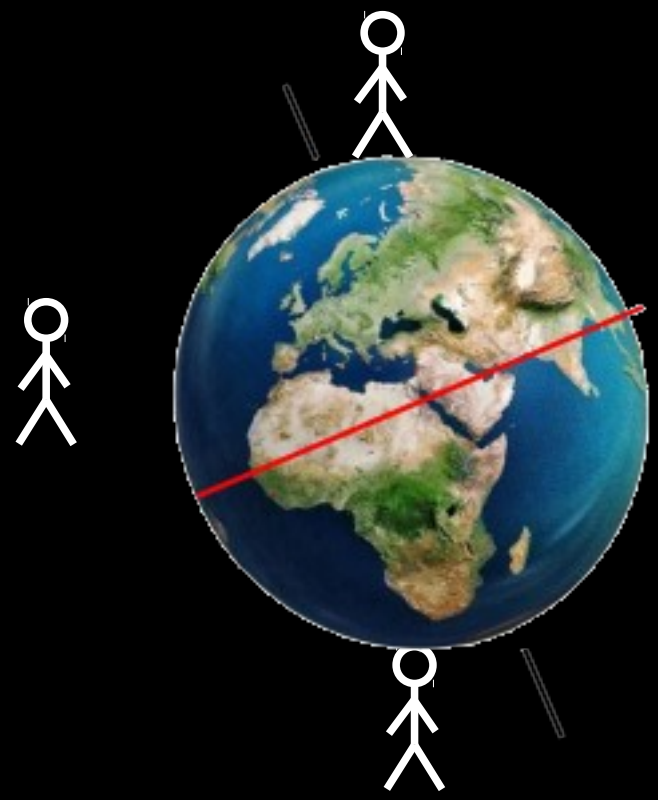


During the lecture(2h)
ca. 30°



Attitude relative to the world is the same
Attitude relative to the floor is the same
But i am rotating! Relativ to what?

Attitude: Angle Between What?

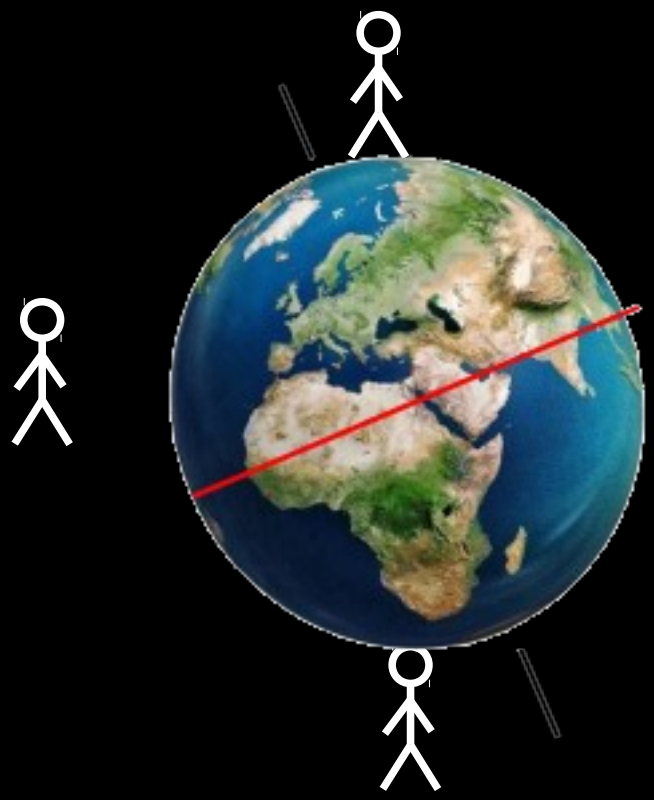


And your attitude? Relative to what?

Attitude: Angle Between What?

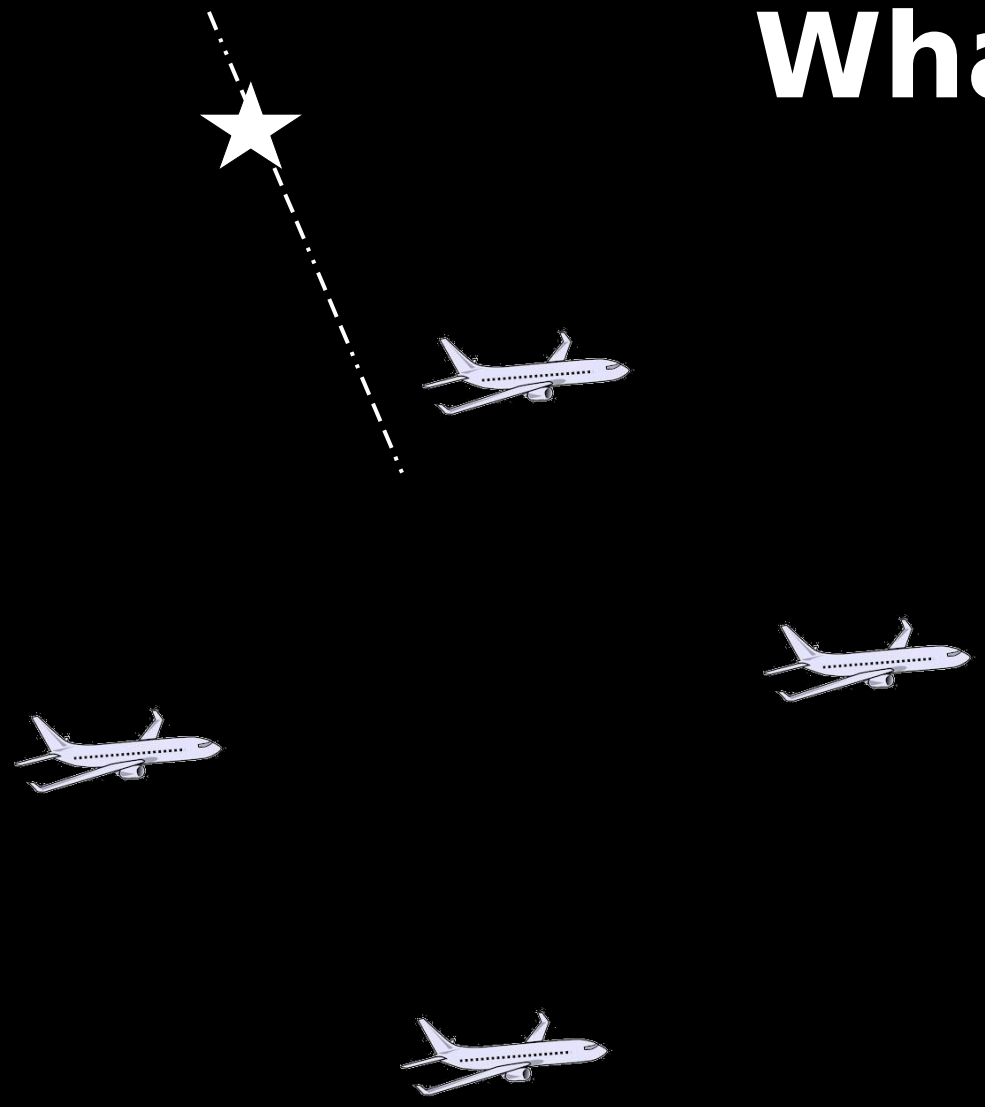
My Attitude?

I'm standing, for sure!
And this indepent from my position!



And your attitude? Relative to what?

Attitude: Angle Between What?



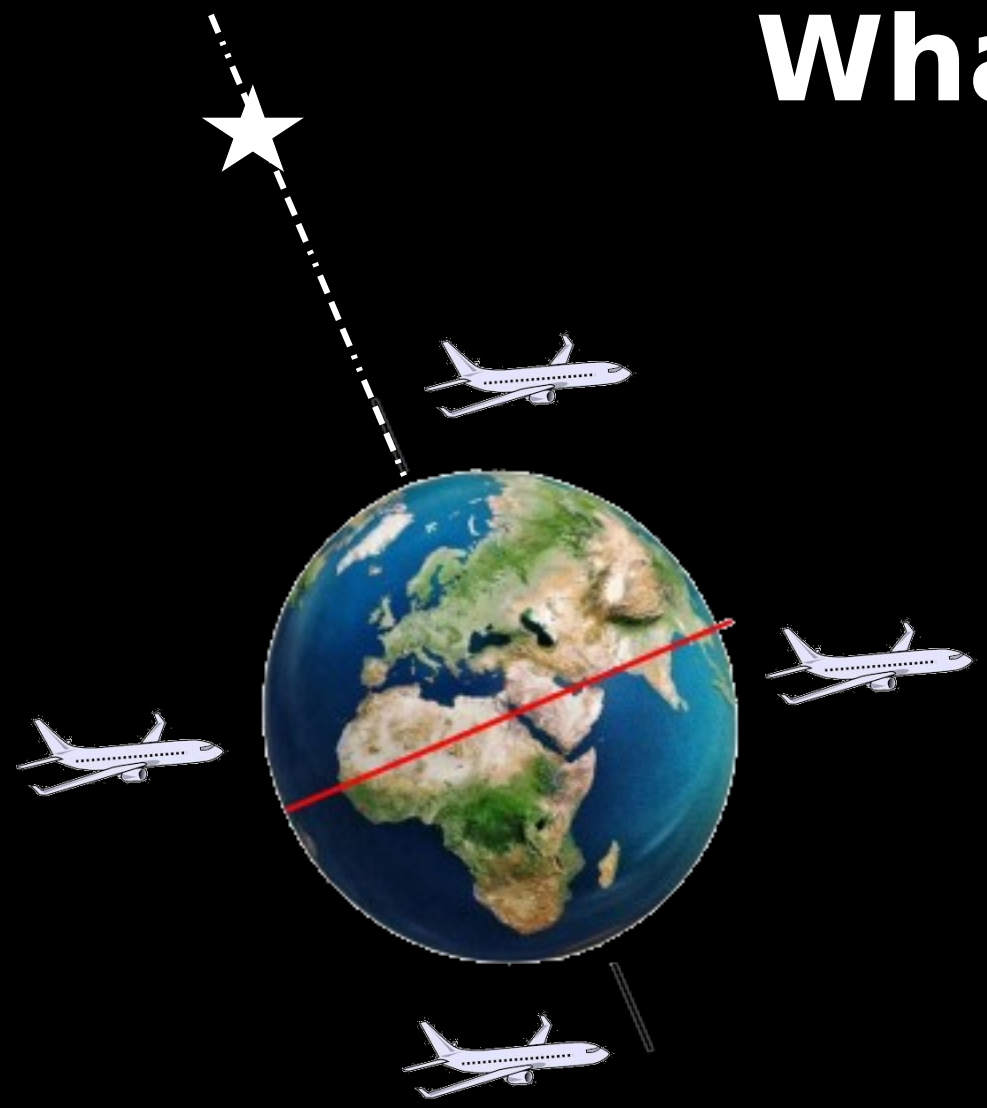
Same Attitude?

- Relative to ground (Earth)
- Relative to stars (eg. polar star / polaris)
- Innertial

Attitude of the earth?

- Relative to polaris
- Relative to Sun

Attitude: Angle Between What?



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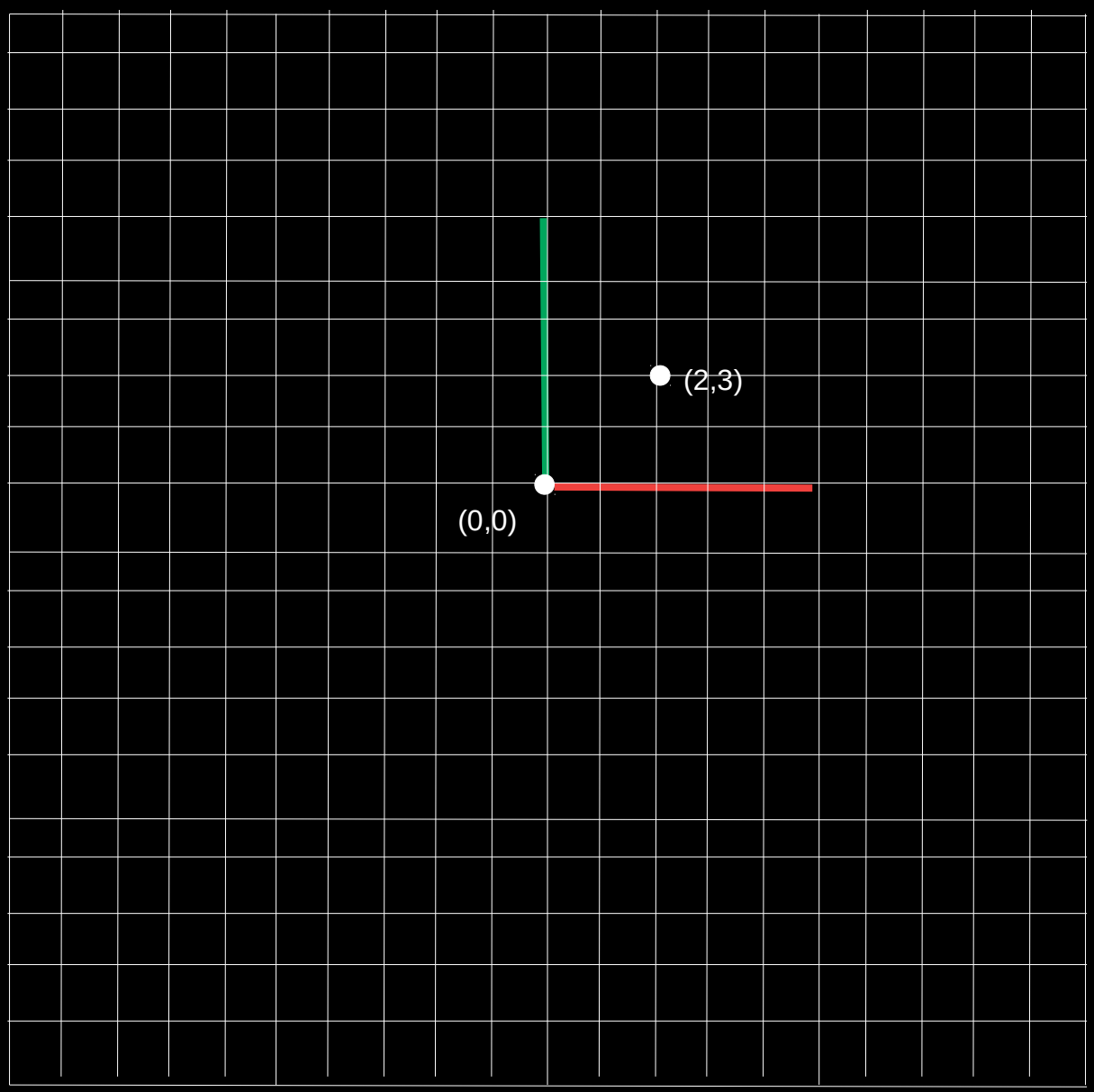
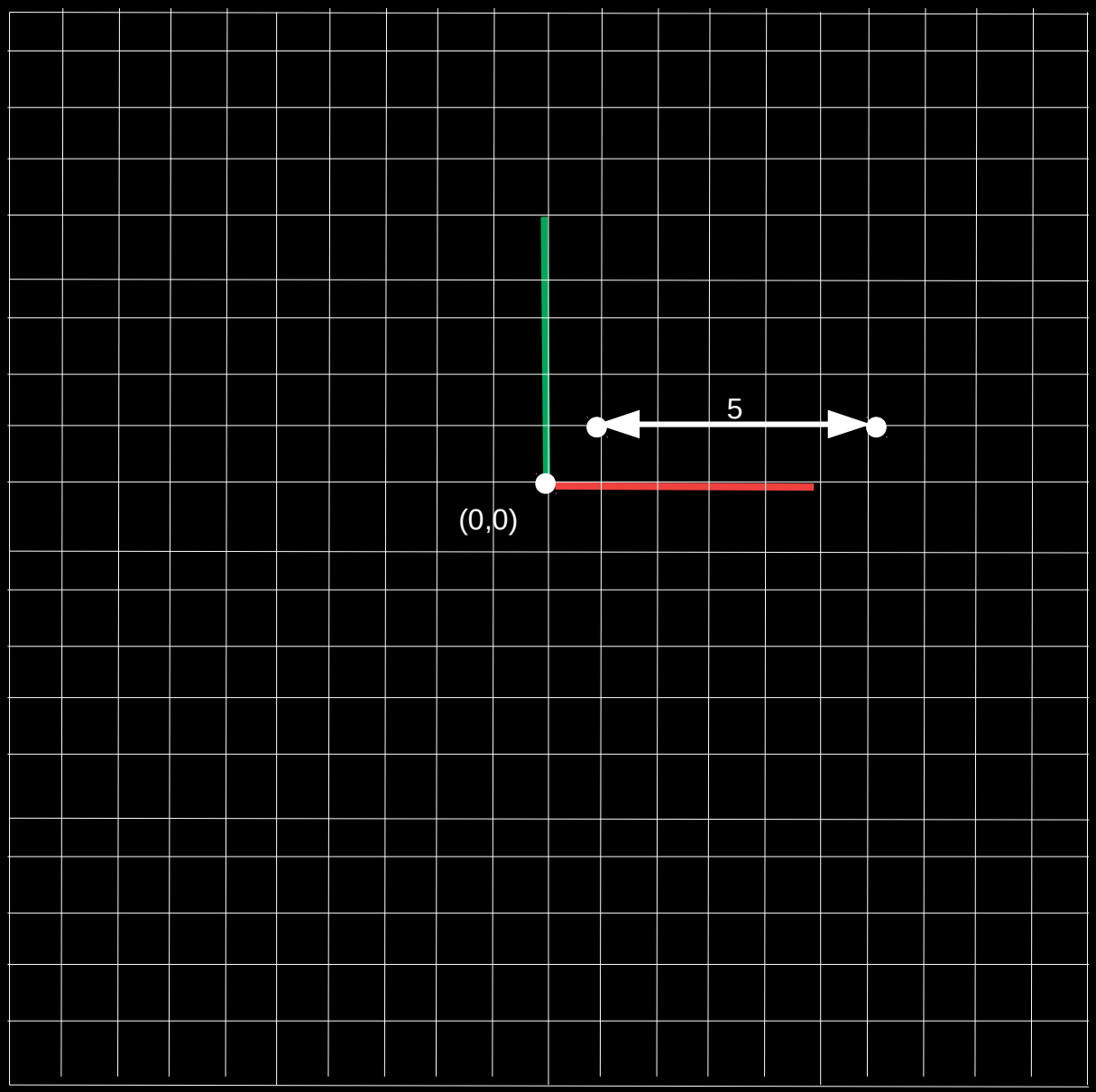
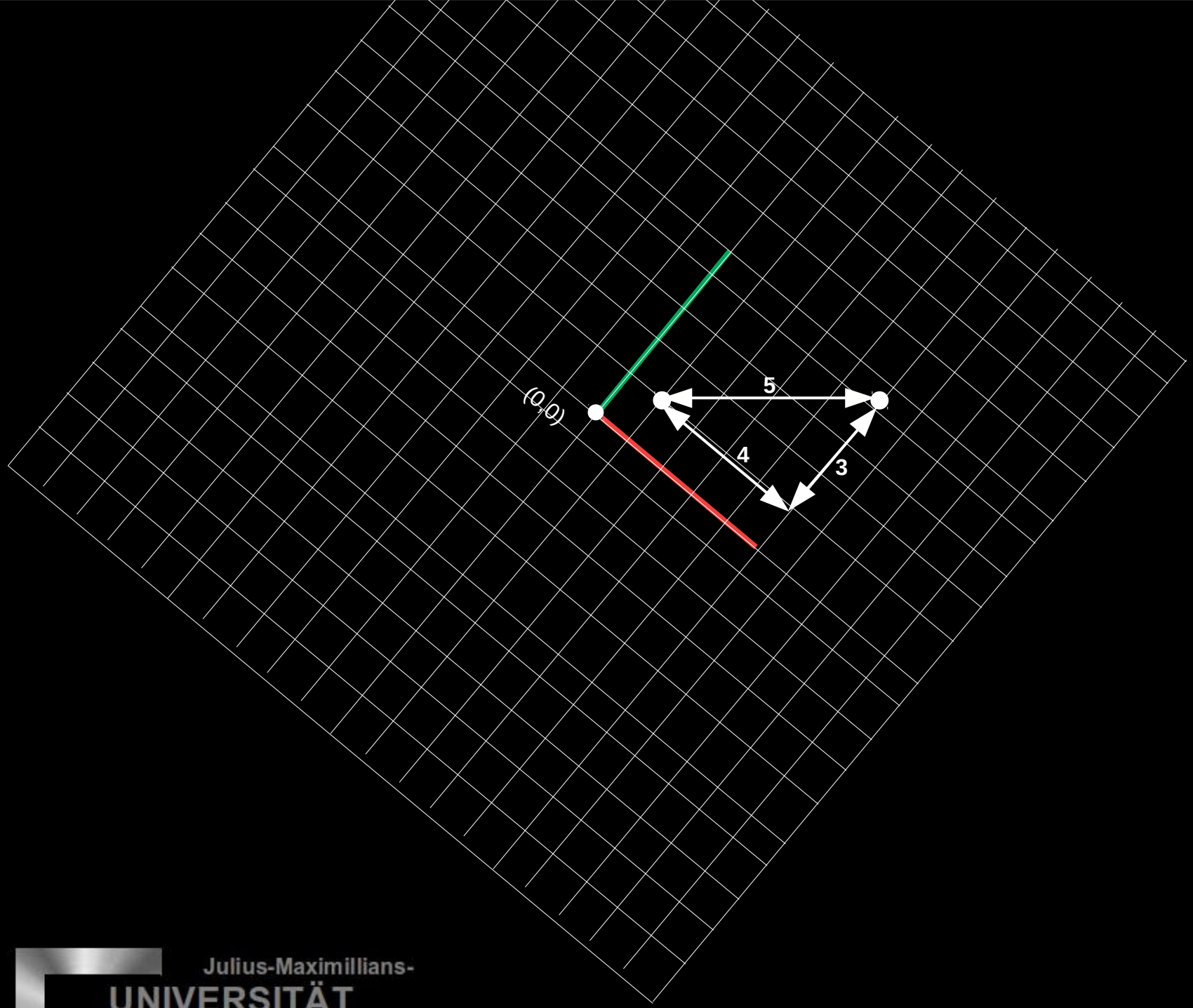
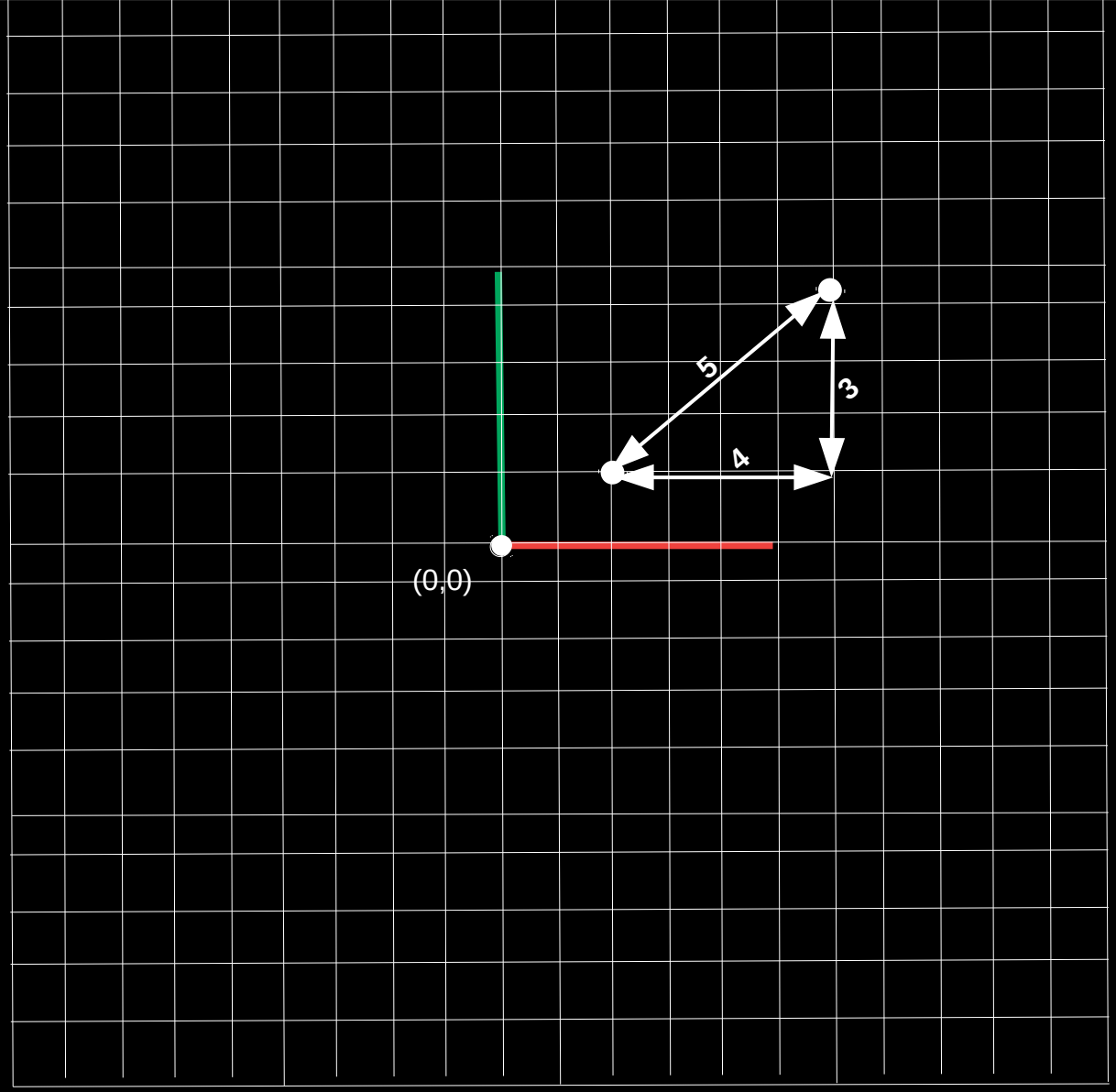

(2,3)
(0,0)

Relative to the slide,
same position.
Relative to frame,
it have moved!



Distance


(0,0)
5
4
3



Distance
stays the same

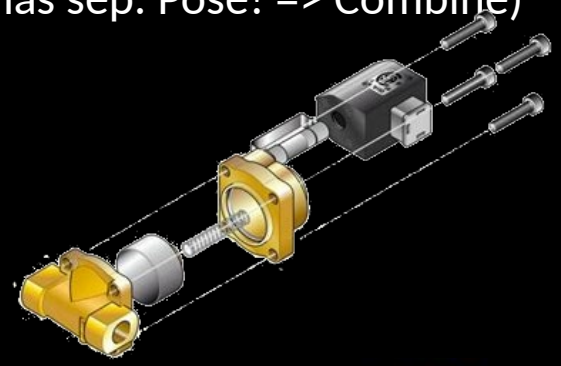
Reference-Coordinate-Systems

How many coordinate systems do we need?

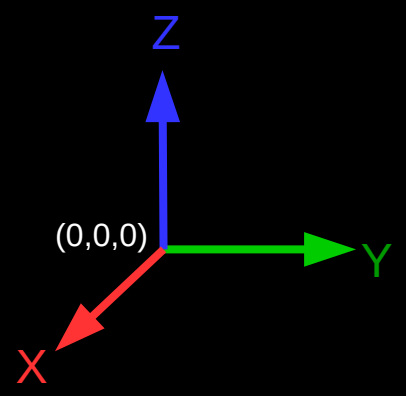
Our position + attitude (=pose), **relative** to:

Earth: Surface (ECEF)	my Office, my flat
Earth: Inertial (ECI)	Orbit
J2000	stars, interplanetary
Laboratory	Orientation to testreceiver
The air around me	Important to keep the plane flying!
Sun, Orbit, Ekliptik , J2000	Interplanetary
Moon	Moon-landing
Built in, relative to structure	**Contruction** (every module has sep. Pose! => Combine)

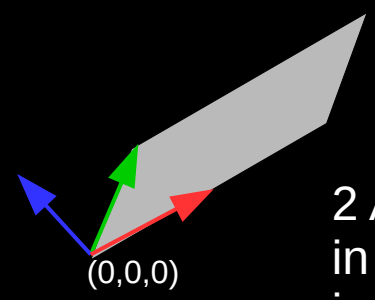
=> Vectors relative to a coordinate system
=> Transformations are of coordinates & vectors



Definition of a Coordinate System



1. Define Origin (0,0,0)
2. Define two orthogonal Axis (A and B)
3. There is no more freedom for the third axis:
   C = AxB
   all Orthogonal & Right-Handed



2 Axes define a surface in space, the third is orthogonal to this surface

Definition of Reference-Coordinate-Systems

Cartesian System

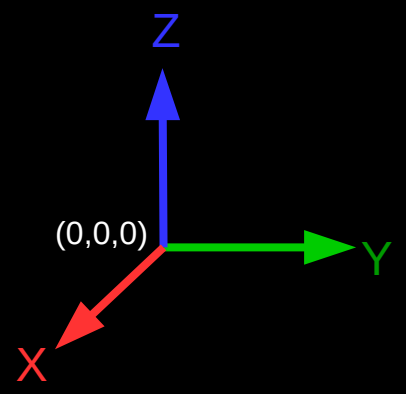
Origin (0,0,0) + 3 orthogonal axis (x-, x+), (y-, y+), (z-, z+)

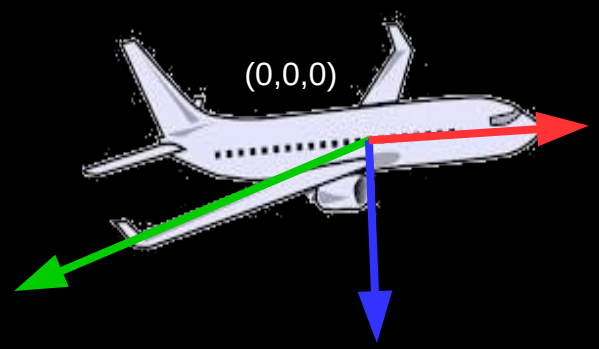
Reference-Coordinate-System:

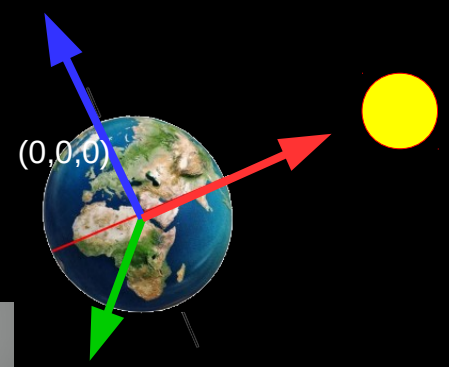
- Origin (0,0,0) … relative to what? And this? No end!
- and orientation (x,y,z) … relative to what?!
  - normal:
    - 3 orthogonal axis
    - Right-Handed
    - Defining of two axis, the third will be derived

X, Y, Z → R$_{ed}$, G$_{reen}$, B$_{lue}$ **XYZ**

Definition of a Coordinate System

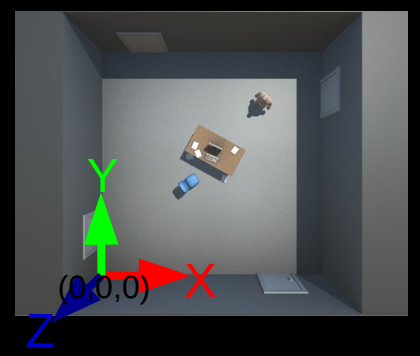




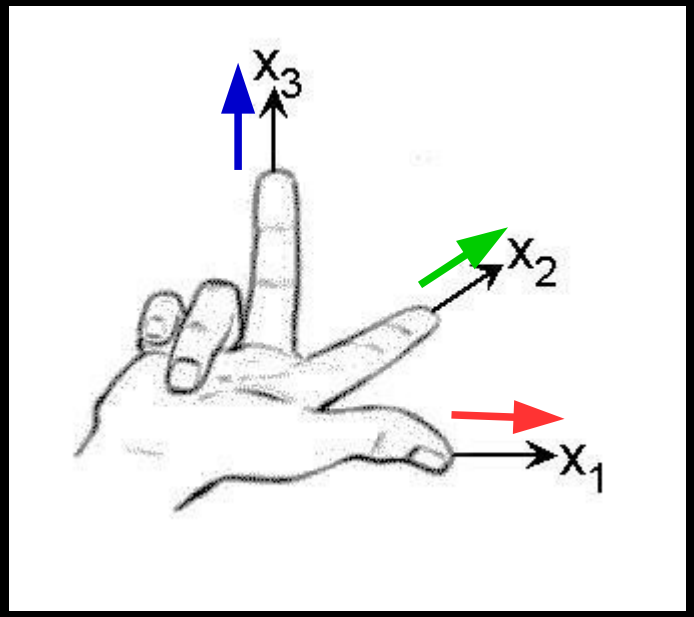


Origin (0,0,0)
and 3 orthogonal Axis

but.....
Relative to what?!?!



Cartesian Coordinate-System: Norm



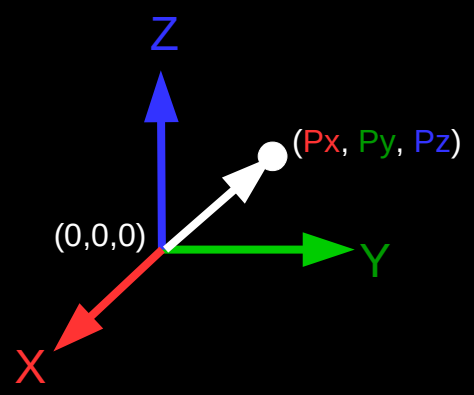
Cartesian Right-Handed
for all Coor.-Sys.

X = x1
Y = x2
Z = x3

Definine: Origin (0,0,0), X-Direction, Y-Direction, Z-Direction
What about front, up, down, left, etc.? => Relative to a extern CS

Nagel Lack... Dagegen kann ich nichts in der Prüfung sagen

Cartesian System



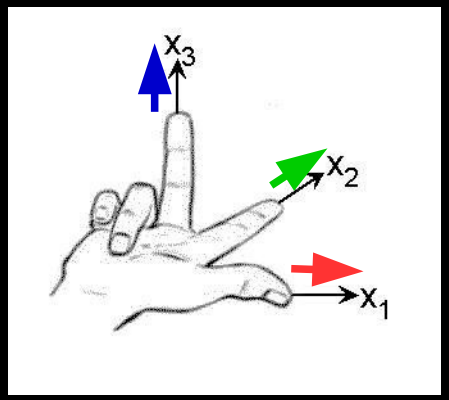
$$\vec{P} = (P_x, P_y, P_z)$$

But a better way: $\vec{P} = \begin{pmatrix} P_x \\ P_y \\ P_z \end{pmatrix}$

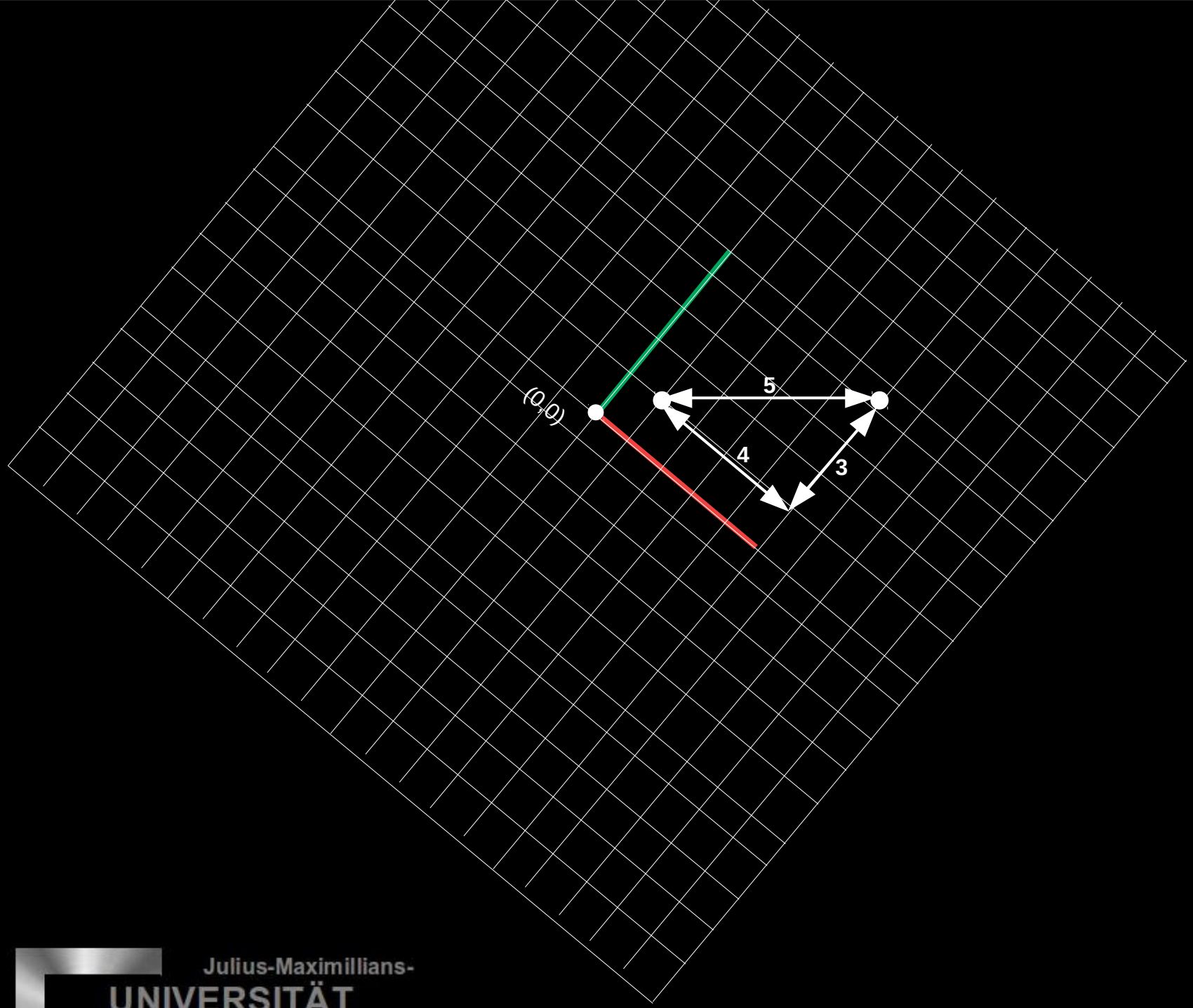
(to multiply matrices)

$$\vec{P} = P_x \vec{e}_x + P_y \vec{e}_y + P_z \vec{e}_z$$

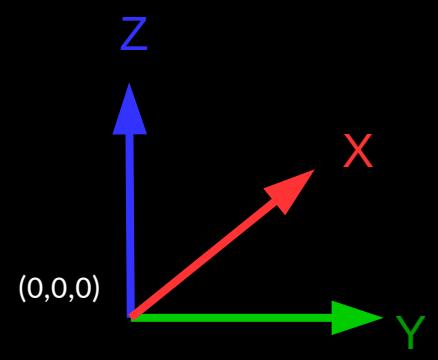
$$\vec{e}_x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \vec{e}_x = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad \vec{e}_x = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$



Ridiculous? Trivial?
But it will helps us with rotations,
They will change!
e_x, e_y, e_z will not remain (1, 0, 0), (0,1,0), (0,0,1)

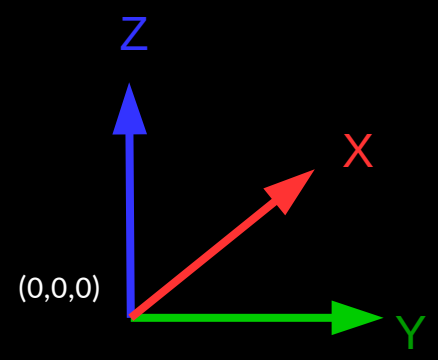

(0,0)
5
4
3

Right / Left

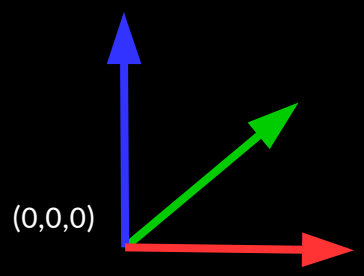


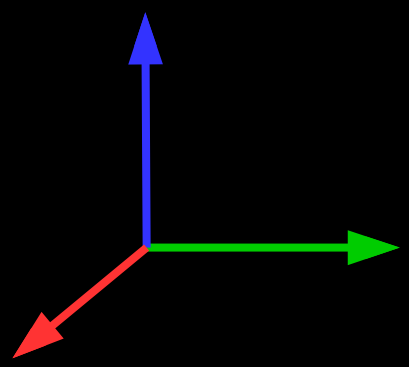
Right-Handed: Right?

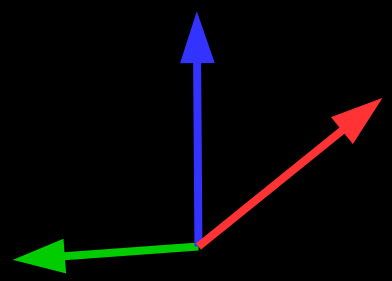
Right / Left



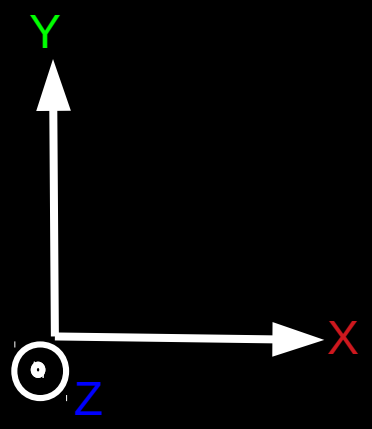
Right-Handed: Right? NO!!
Either: exchange two axis/
invert one axis



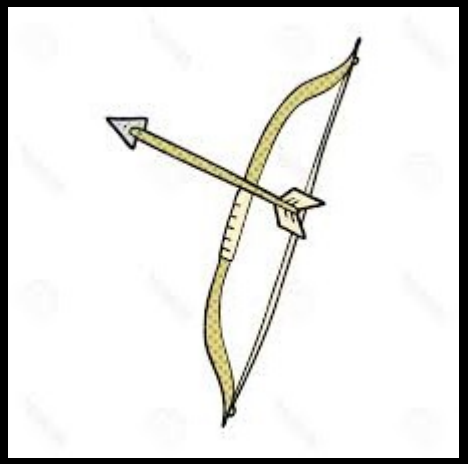




Flat Drawing



Right-Handed



⊙ Vector points to me

⊗ Vector points away from me

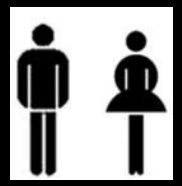
Reference-Coordinate-Systems

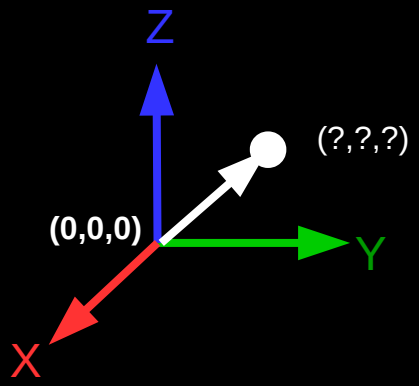
Reference (0,0,0)?

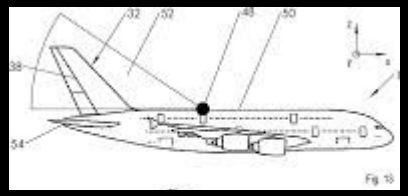
- Center of the earth
- Belly button
- Tip of plane
- Plane's Center of Gravity (COG)
- Sun
- Center of the universe

What else?

Me! (and when I'm moving?)







Many different! We must be able to convert from every system in another one
From which to which? An „absolute“ (not relative) system would be nice!

...or a central Reference-Coordinate-Systems
and any other will be converted

Important Reference-Coord.-Sys.
(Frame = Ref.-Koord.-Sys.)

Some more examples....
3 orthogonal axis (x-, x+), (y-, y+), (z-, z+) ...Do they have to be 100% orthogonal?

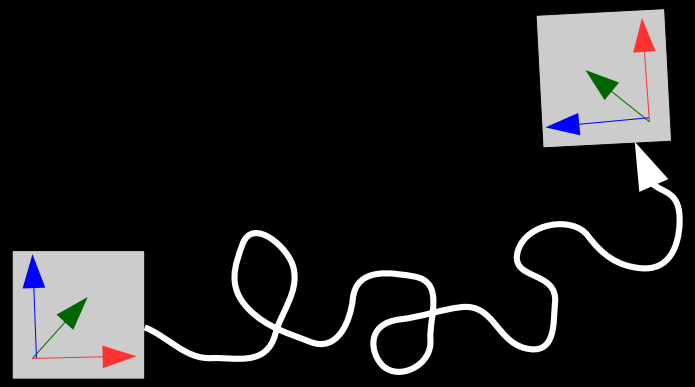
- Body
  - Structural Coordinate Frame SCF
  - Mechanical Coordinate Frame MCF
  - Principal Axis-Body-Fixed Frame BF
- Earth
  - Earth-Centered Fixed Frame ECEF or WGS
  - Earth-Centered Inertial Frame ECI
    - True (Equinox and Equator) of Date TOD
    - International Celestial Reference Frame ICRF
- Orbit
  - Local Vertical Local Horizontal LVLH
  - Celestial Coordinate Systems CCS

Many different! We must be able to convert from every system in another one!
Translation, rotation and all as a function of time (it moves!)

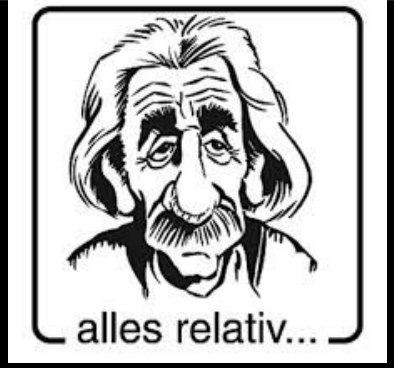
Two Special Frames

1. Inertial Independent of the universe

2. Δ Pose: Relative to my former Position of xxx
   e.g. J2000 Frame (Earth position on 1/1/2000 at 0:00)



Inertialsystem



Every thing?

A very special Reference-System
independent of the universe

position, attitude, v, a, ω

Inertial: accelerationsensor + rotationsensor
Acceleromenter Gyro

It only provides
angular velocity
&
acceleration

Is that Δ Pose?
=> No! Δ Pose is (v * time), we do not have "v" only acceleration

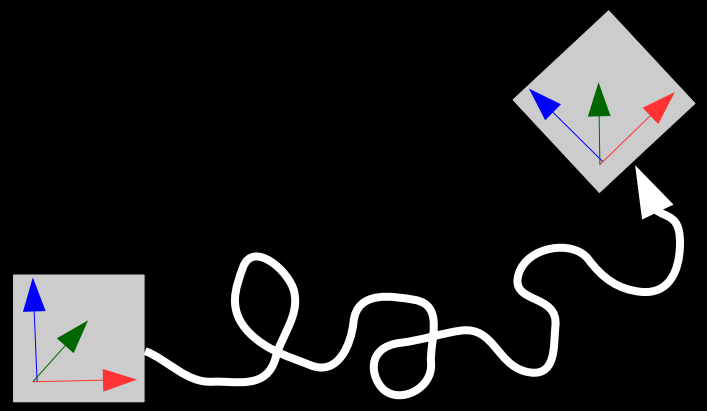
Independent of the universe and matter (Absolute?)

then... movements relative to what?

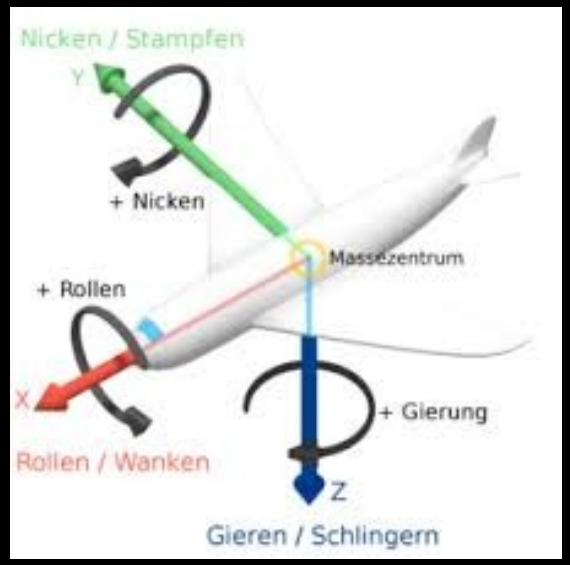
Δ Pose

A very special Reference-System
independent of the universe

Δ Pose: Relative to an old Position of xxx
e.g. J2000 Frame (Earth: Position am 1/1/2000 um 0:00)
Independent of the rest of the universe
but than...
How do I measure?=>**Inertial**
In which direction did I move?

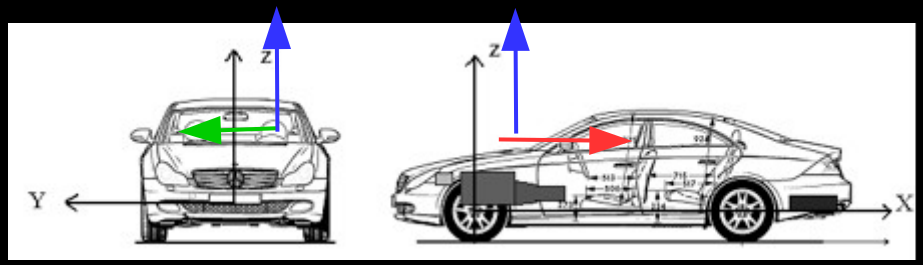


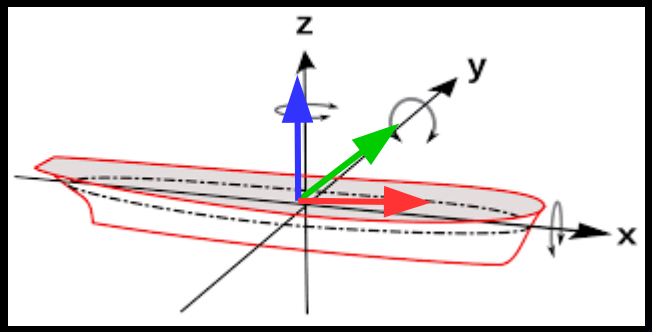
„Our“ Examples (all identically consistent?)



Def. +/- angles of a plane
056a-rotationen.odp ab 15

Attention! Satellites have no standard für Body-Coordinate-Sys.





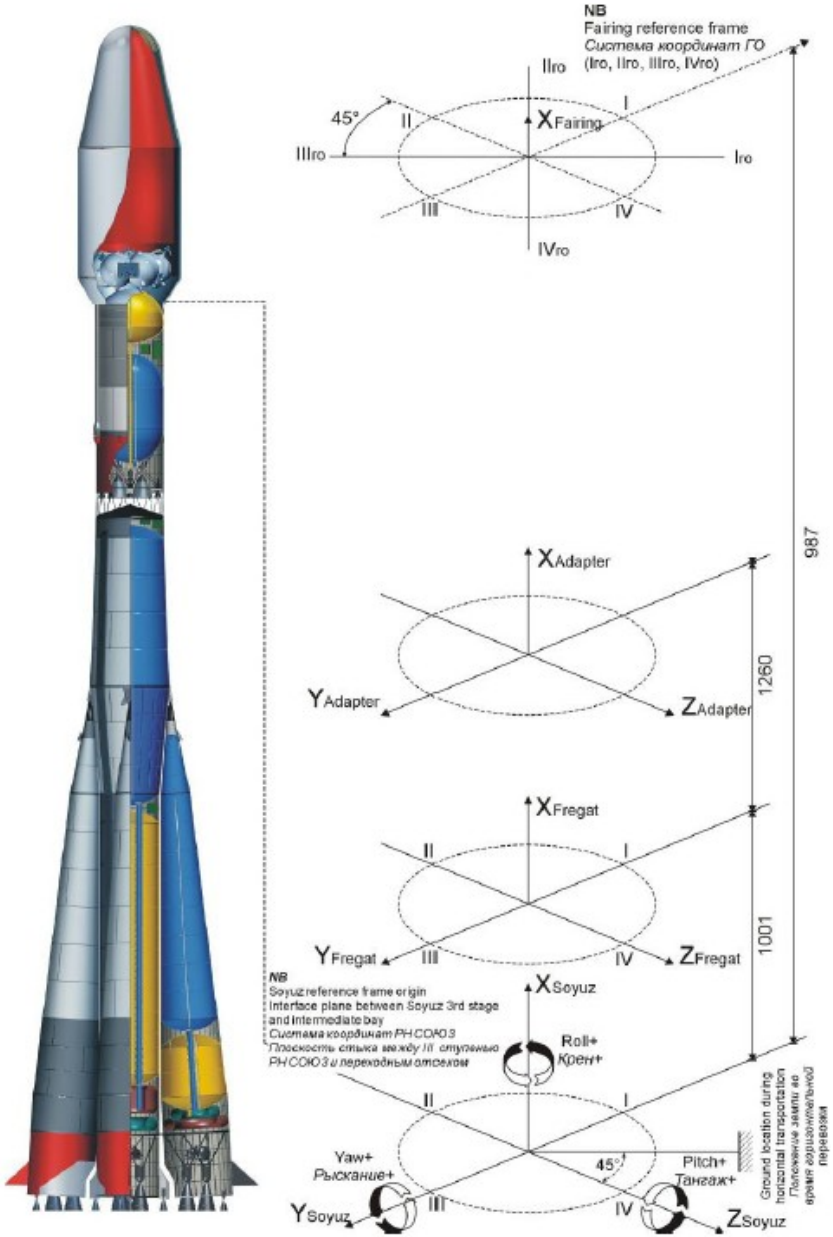
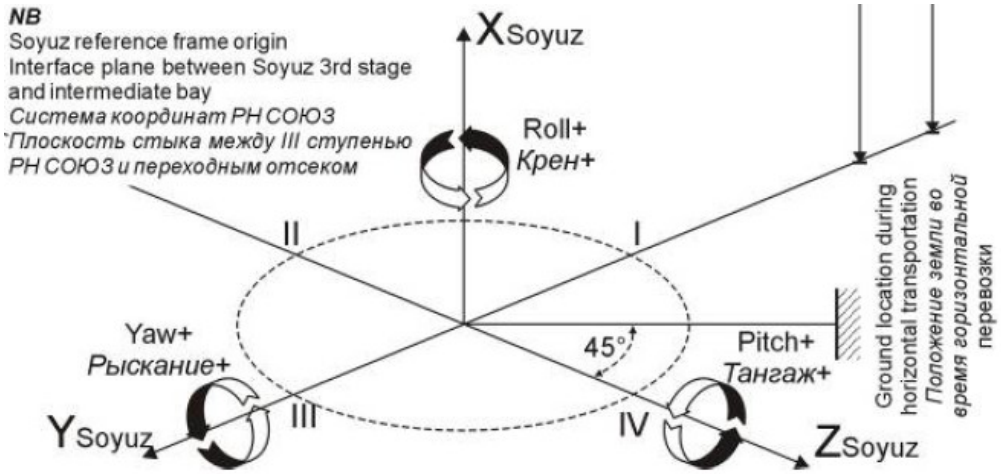


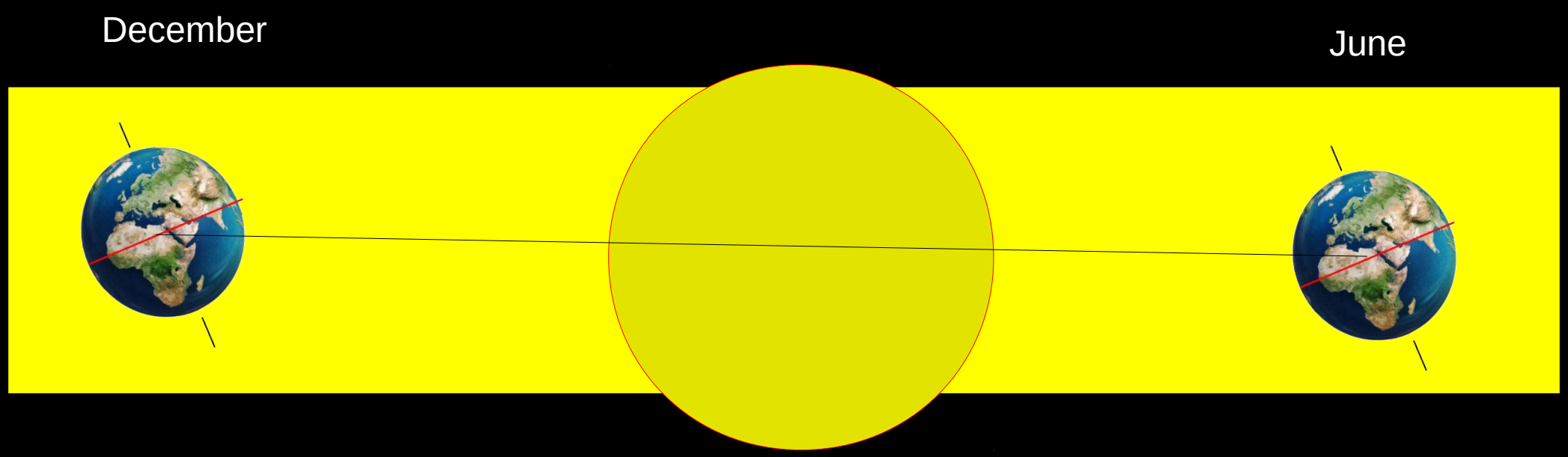
Figure 7-1: General coordinate system of the Launcher [LV RD 2]



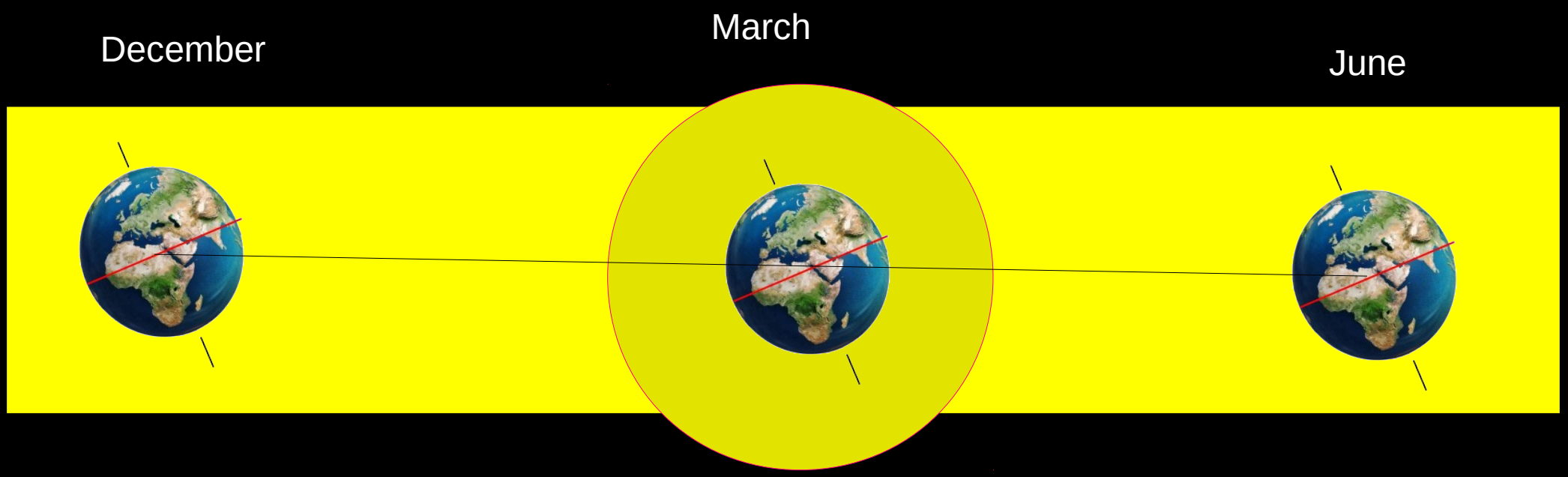
X heads forward!

Refer slide 66 (planes)

Equinox



Equinox



Equinox and Earth-Inertial

UT date and time of equinoxes and solstices on Earth[1][2]

event	equinox		solstice		equinox		solstice	
month	March		June		September		December	
year	day	time	day	time	day	time	day	time
2016	20	04:31	20	22:35	22	14:21	21	10:45
2017	20	10:29	21	04:25	22	20:02	21	16:29
2018	20	16:15	21	10:07	23	01:54	21	22:22
2019	20	21:58	21	15:54	23	07:50	22	04:19
2020	20	03:50	20	21:43	22	13:31	21	10:03
2021	20	09:37	21	03:32	22	19:21	21	15:59
2022	20	15:33	21	09:14	23	01:04	21	21:48
2023	20	21:25	21	14:58	23	06:50	22	03:28
2024	20	03:07	20	20:51	22	12:44	21	09:20
2025	20	09:02	21	02:42	22	18:20	21	15:03
2026	20	14:46	21	08:25	23	00:06	21	20:50

Axis fixed like:

in 20th March 2000,

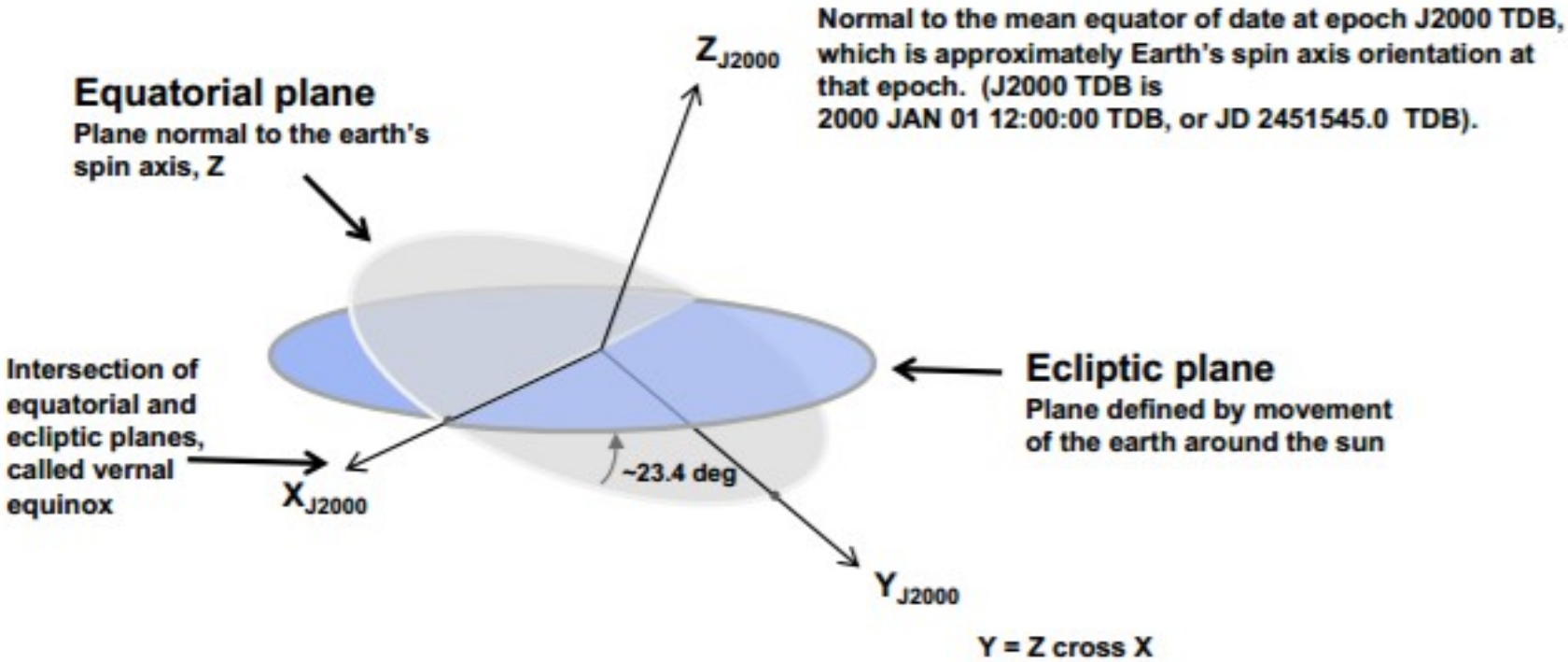
at 7:35 (UTC)

Origin in the earth's center of gravity and is moving with it

Where was it?

J2000 Frame

The J2000* frame definition is based on the earth's equator and equinox, determined from observations of planetary motions, plus other data.



Examples of Reference-Coord.-Sys.

central body reference frames.
- J2000
- FixedAtJ2000
- MOJ2000
- TOJ2000
- TEMEJ2000
- TETEGM
- MOD
- TOD
- TEMED
- PseudoFixed
- Fixed_NoLibration
- PrincipalAxes_403
- PrincipalAxes_421
- PrincipalAxes_430

Trajectory Based Reference Frames
- ICR
- LVLH
- RIC
- VNC
- VVLH

Convert:
Translation
Rotation
but...
They are moving!
=> f(time)

Epoch-based Reference Frames
- AlignmentAtEpoch
- MeanOfEpoch
- TrueOfEpoch
- TEMEOfEpoch

Ecliptic Based Reference Frames
- MeanEclpDate
- MeanEclpJ2000
- TrueEclpDate
- TrueEclpJ2000

Central Body Surface Based Reference Frames
- NorthEastDown
- NorthWestUp
- RadialCentric
- TopoCentric

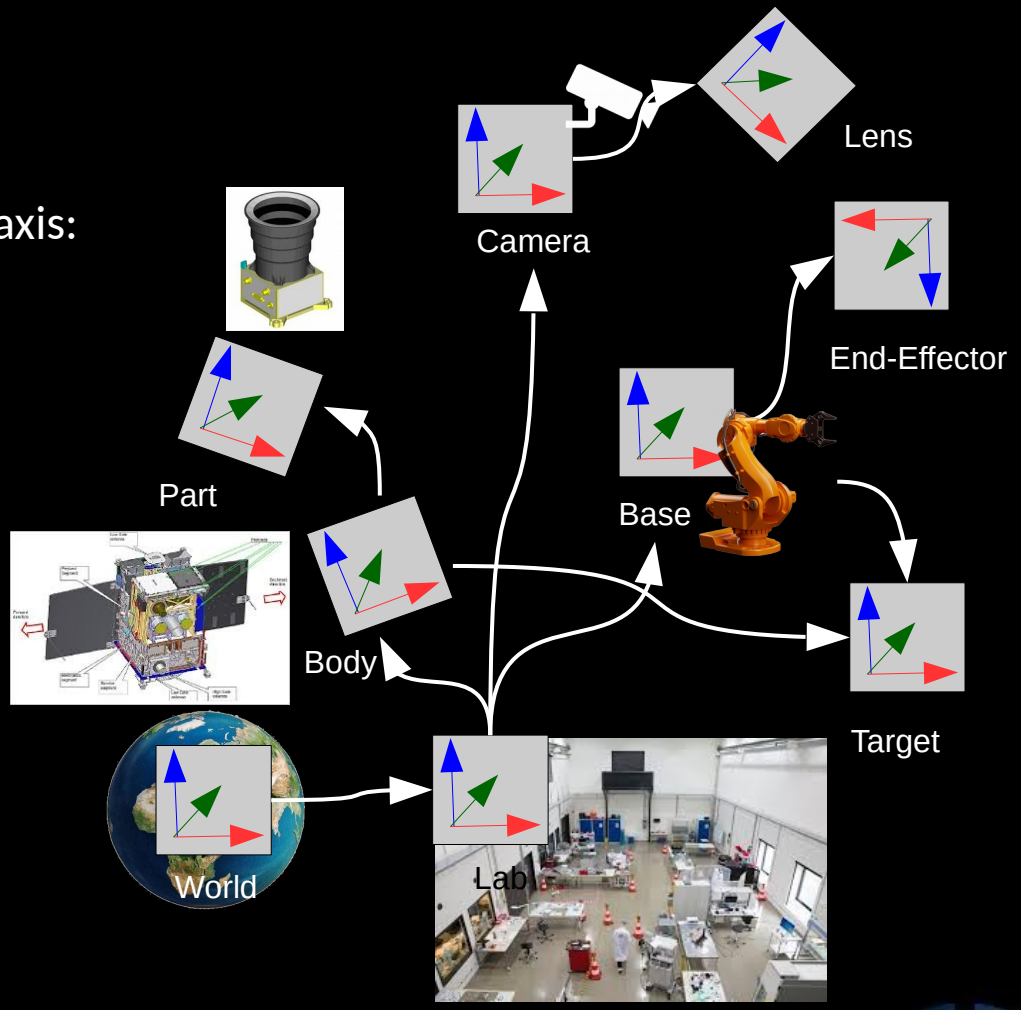
http://help.agi.com/stk/index.htm#stk/referenceframesvgt.htm

Definition of the Reference-Coordinate-System

1. Define Origin (0,0,0)
2. Define two orthogonal Axis
3. There is no more freedom for the third axis:
   Orthogonal & Right-Handed

Mapping from every to any system:
Translation + Rotation: Pose
probably (most certainly!) f(time)

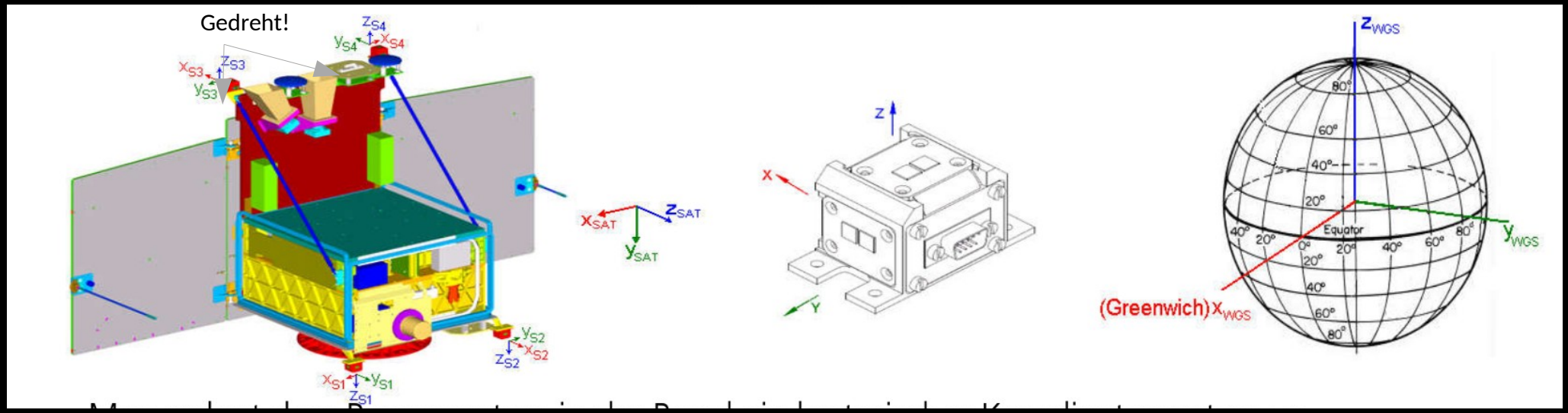
„**Mounting Frames**“



Mounting Frames

Eg. Direction of the Sun, from each Sun-Sensor

Every photosensor
gives „the“ sunvector



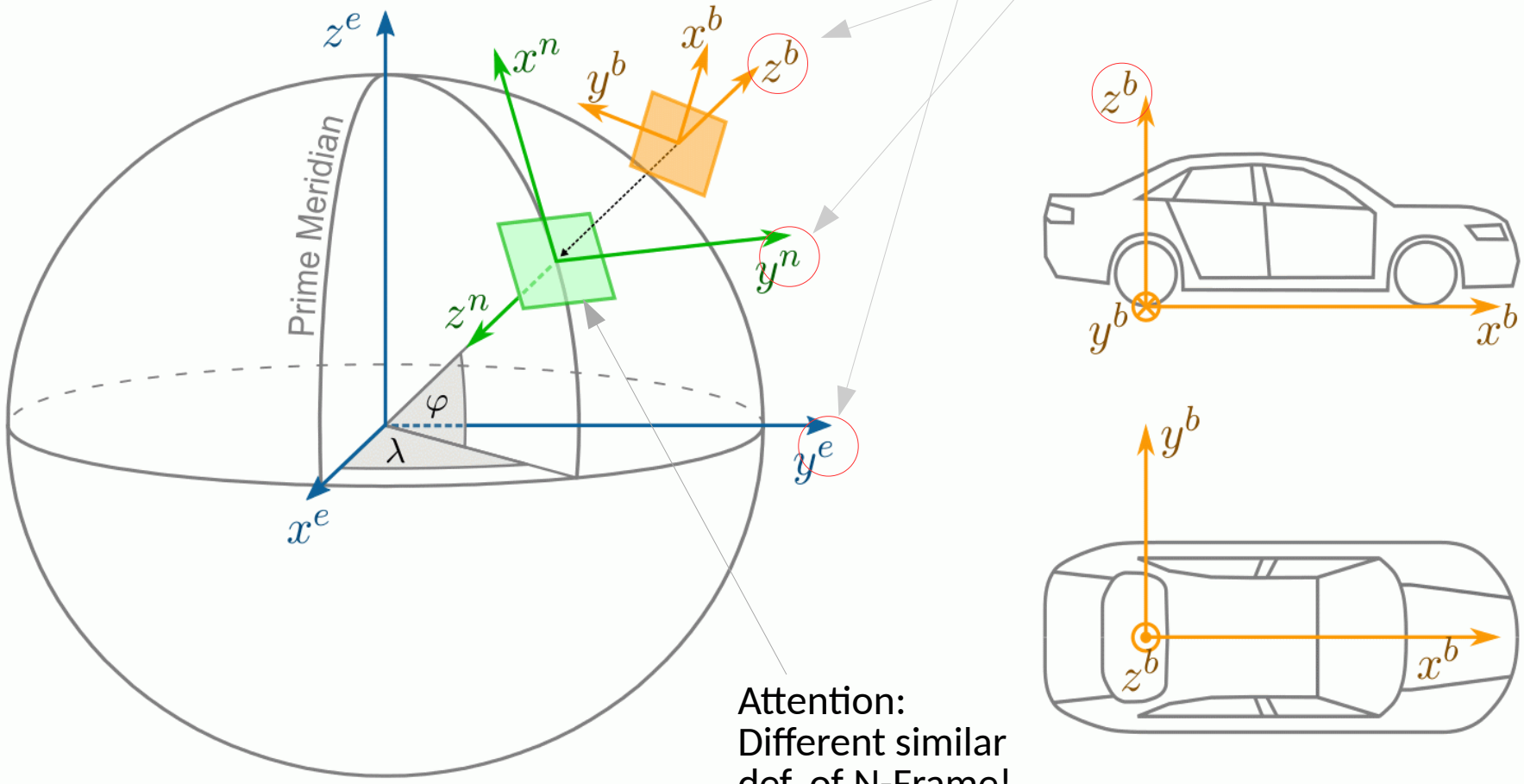
Reference-Coordinate-Systems
(Frame = Ref.-Koord.-Sys.)

Inertial	IMU, Rotations & Acceleration (!= **Δ Pose**) Independent of the universe and matter (Absolute?)
Body anchored	
Structural	Construction
Mechanical	Dynamic, Rotation, Translation
Earth	
ECEF: earth surface	Adresses & Co
ECI: Inertial	Orbit, Intercontinental Flights
Arial Navigation Frames	
E	Relative to **E**arth surface (ECEF)
I	IMU, **I**nertial,
B	Mechanical (**B**ody)
N	Orientation on earth (**N**avigation) (Question N == E?)

Position of a frame relative to another one?

3D Translation & 3D Rotation… perhaps several times

Frames (= Ref.-Coord.Sys.)
Mouting Frames: e, n, b



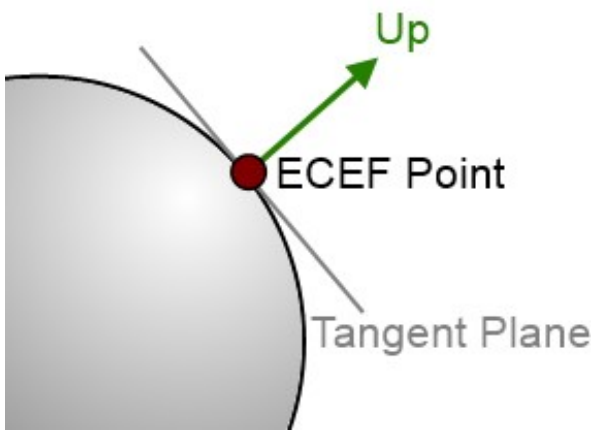
Attention:
Different similar
def. of N-Frame!

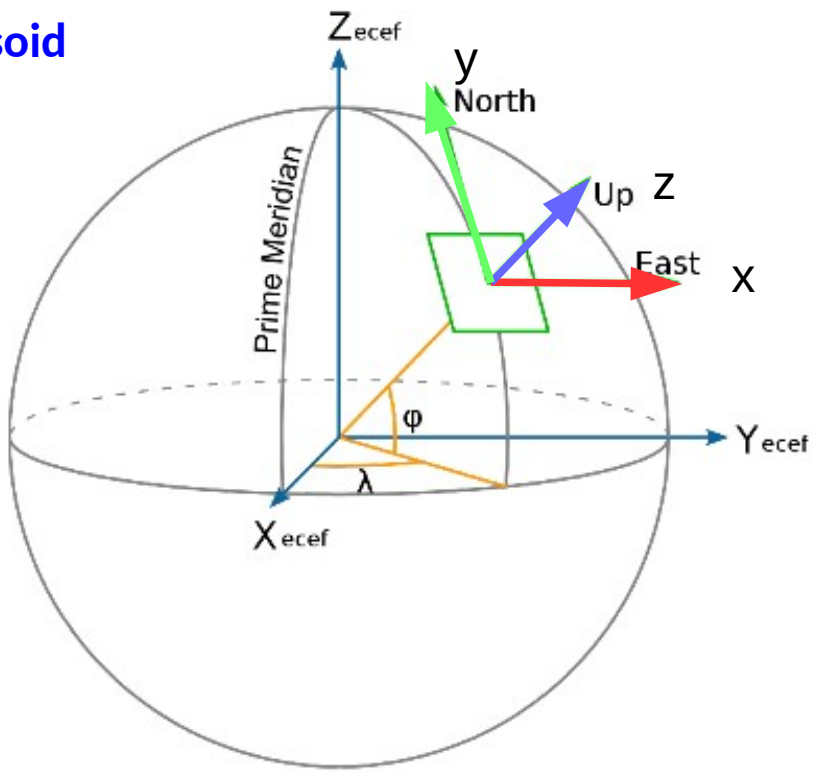
N-Frame: Orientation

Same origin as the b-Frame (Mechanical)

y-axis is pointing North (rotational axis)
x-axis is pointing East
both axis are in a **tangential plane to the WGS 84 ellipsoid**

z-axis is pointing up, parallel to the gravity.





Parameter on Mounting Frames

E	Relative to Earth surface (ECEF)
I	IMU, Inertial,
B	Mechanical (Body)
N	Orientation on earth (Navigation)

And relative to the surrounding air?

All parameters:

$$V^{n}$$

v = parameter, e.g. velocity
n = Reference-Frame (CS)

Parameter on Mounting Frames

These mounting parameters are used for

- Attitude Control,
- Guidance,
- Navigation,
- Payload (Telescopes, Sensors, etc.)

Devices have a defined orientation within the satellite
It is given by individual **mounting frames** and shall be documented

Transform V von a => b: V^{a}_{b} or V_{ab}

$V_{eb} = V_{ei}$ thenn V_{ib} eg. V = Speed

from e to b from e to i and then from i to b

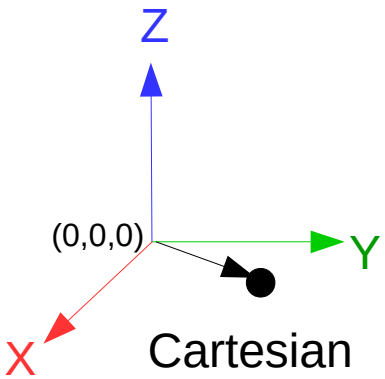
Julius-Maximillians-
UNIVERSITÄT
WÜRZBURG

Aerospace Information Technology

Just for Information: Polar & co

Which one is the best (for whom) and why?
What for? (Translation or Rotation or both?)

Easy for translation & distance
Terrible for rotations

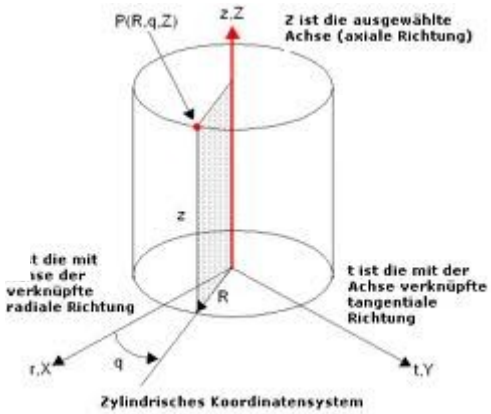


Cartesian

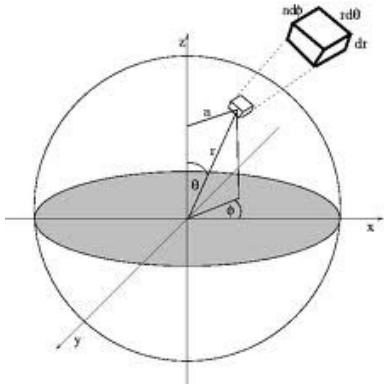
Our whole mathematics based on the cartesian CS:

So, all calculations we do in th CS
... even if it is sometimes harder to do

No meaning for us

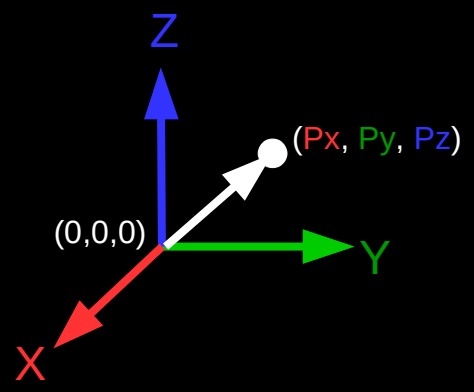


Easy for rotations
Terrible for translation & distance



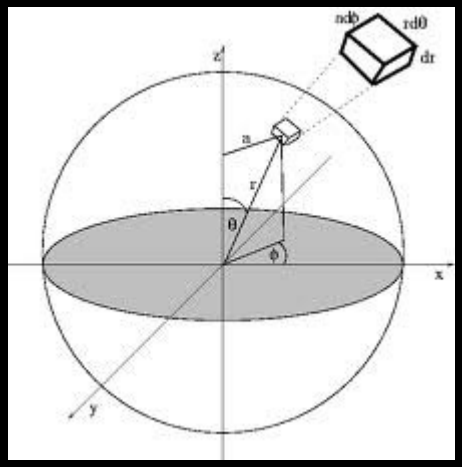
Positions on earth

Translation & Rotation



Transformation

many sin, cos, +, -, *, /



I calculate translations like this:
(just + & -)

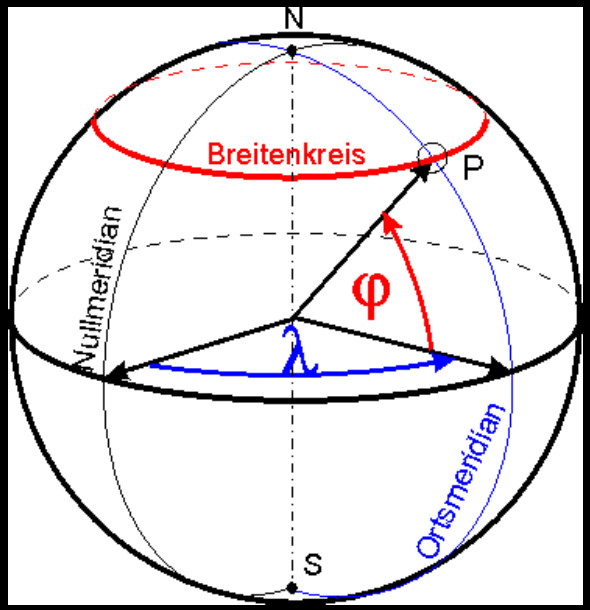
I calculate rotations like this:
(just + & -)

Did I win something by doing this?
Only confusion!

=> Let's stick to the cartesian

Polar-Coordinate-System
Coordinate on Earth

(GNSS, e.g. GPS calculates everything based on cartesian Coordinates)



Polar-Coordinate-System
Longitude
Latitude

Only 2 parameters?
There should be 3!

Why not cartesian?

- And how do we think of translations < 100 km?
- Distance point A … point B?
- Measurements, property…
- Refer „e-“ and „n-“ frames

Earth: Where ist North? Magnetic, rotation axis, perpendicular to Ecliptic, …?

Body Anchored Coordinate Systems

Coordinate Systems : SCF

Structural Coordinate Frame (SCF)
The SCF is a spacecraft body fixed system with its origin located in the center of the bottom plate of the spacecraft where the launcher adapter is mounted

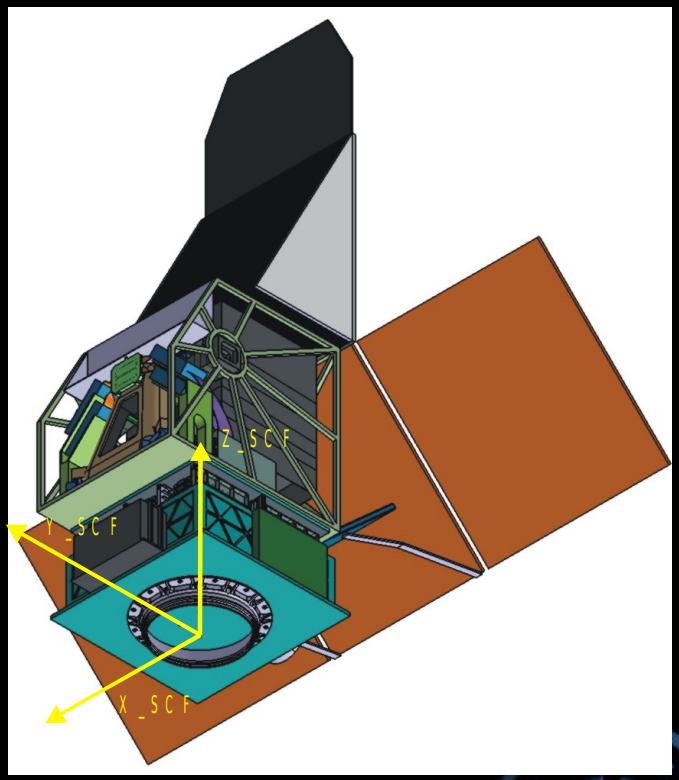
Example of definitions:

Origin: Center of the bottom plate of the spacecraft where the launcher adapter is mounted

$-Y_{SCF}$ perpendicular to the body fixed solar panel and is pointing “outwards”

$+Z_{SCF}$ is perpendicular to the launcher adapter and is pointing from the geometric center into the direction of the payload segment

$+X_{SCF}$ building a **right handed** orthogonal system



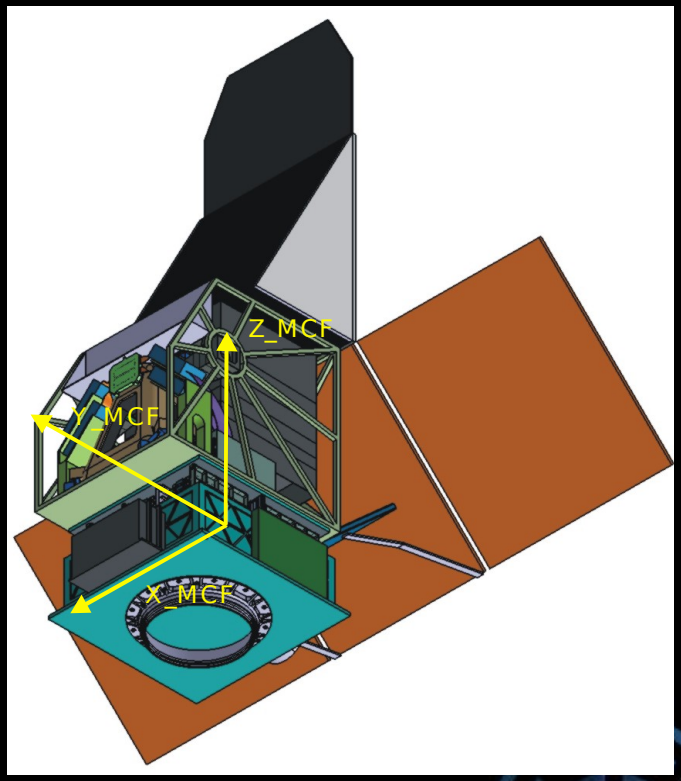
Good to build...
but bad for rotations

Coordinate Systems : MCF

Mechanical Coordinate Frame (MCF)
Spacecraft body fixed system with its axes oriented parallel to the SCF axes.
Origin lies in the Center of Mass (COM) of the spacecraft.
Warning: Center of Mass will change during design and maybe also during mission lifetime (deployable parts or consumables).

Example of definitions:

Origin:	Spacecraft Center of Mass (before or after deploy?)
$+X_{MCF}$	parallel to $+X_{SCF}$
$+Y_{MCF}$	parallel to $+Y_{SCF}$
$+Z_{MCF}$	parallel to $+Z_{SCF}$



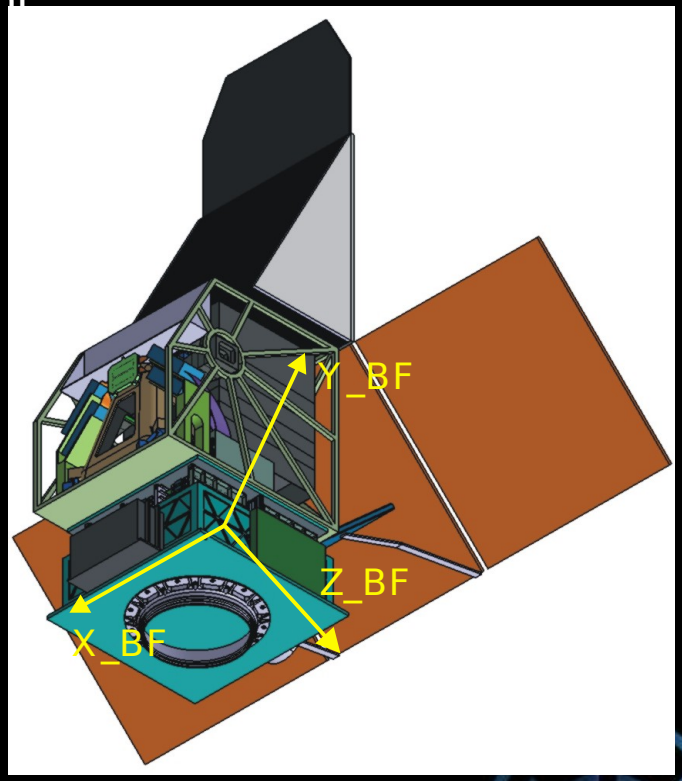
Transformation SCF ↔ MCF ? => Only translation

Coordinate Systems : BF

Principal Axes Body Fixed Frame (BF)

e.g. tumbling motion

Oriented to moment of inertia
The moment of inertia tensor is a real-symmetric matrix.
Three orthogonal eigenvectors and eigenvalues.
The three eigenvalues are the principal moments of inertia.
Center of mass and principal axes will change during design and maybe also during mission life time: AOCS must know this development



Example of Definitions:

Origin: Spacecraft Center of Mass

$+X_{BF}$ Principal Axis closest to X_{SCF}

$+Y_{BF}$ Principal Axis building a right handed system

$+Z_{BF}$ Principal Axis building a right handed system

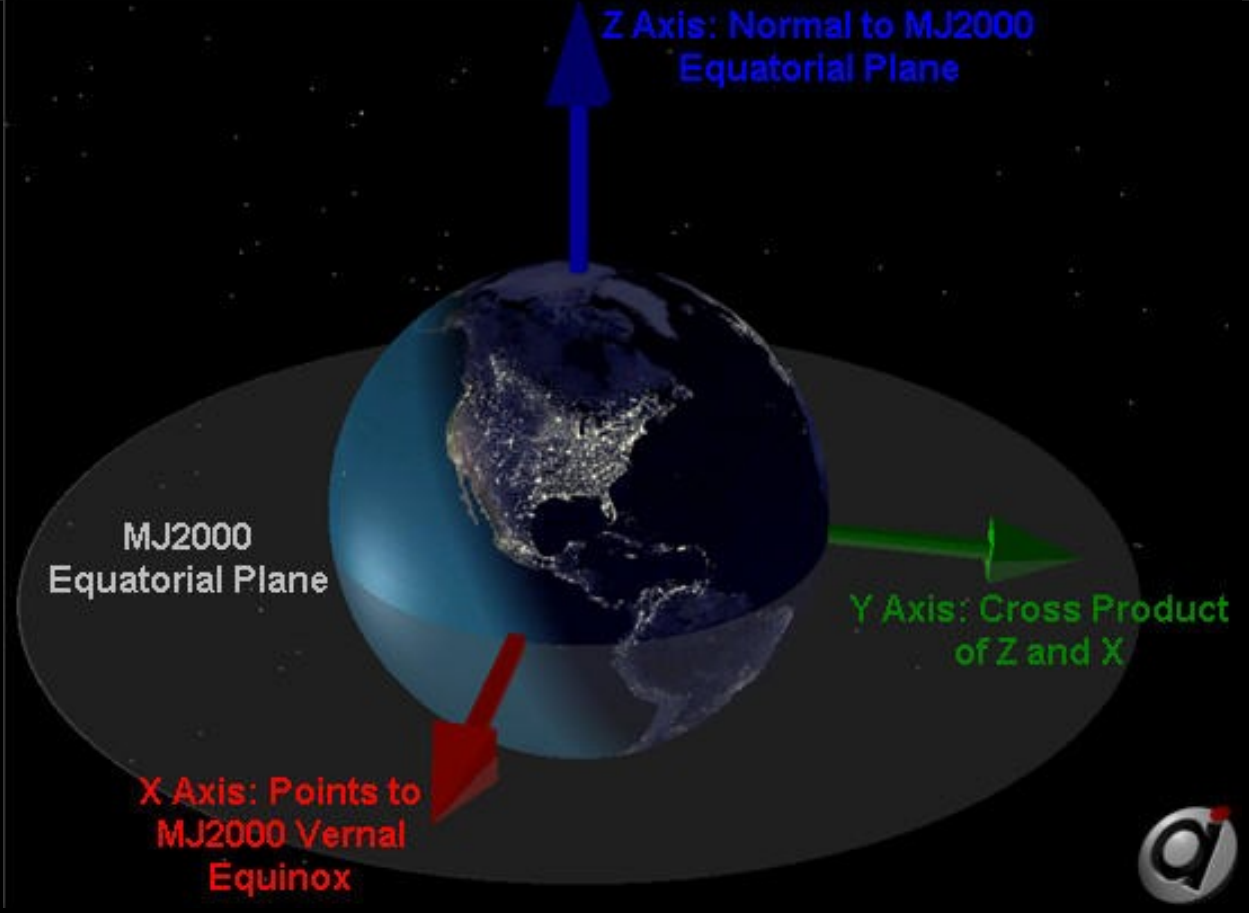
Transformation SCF ↔ MCF ? Translation and rotation
(determined matrix?)

Coordinate Systems on Earth
ECEF & ECI
ECEF <=> ECI : f(t)

Siehe 055b-ecef-eci.odp

There are many different...
e.g.
Celestial Stars

https://ai-solutions.com/_help_Files/attitude_reference_frames.htm

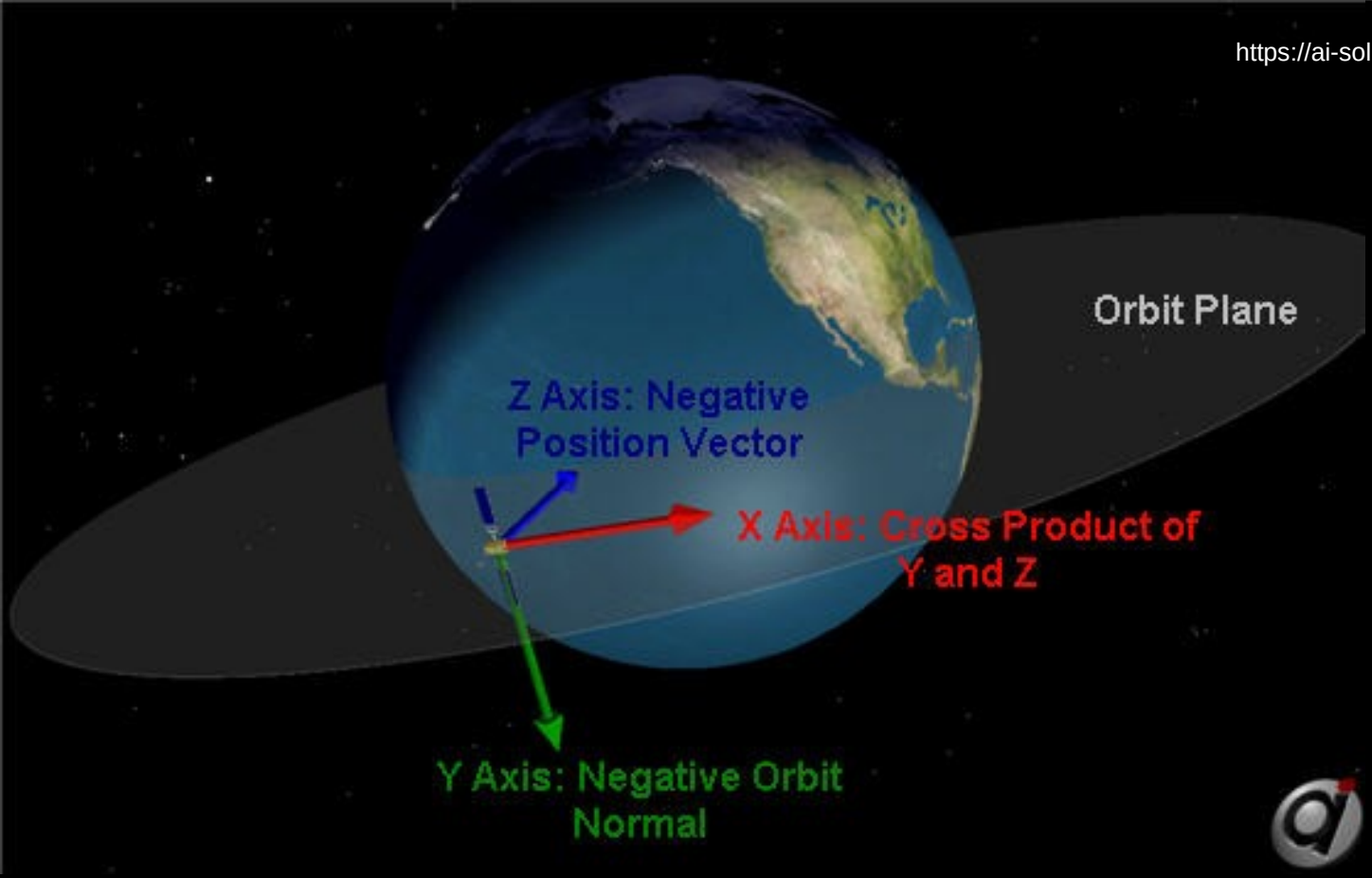


MJ2000 Earth Equator

Description:
•Z-axis: Vector normal to the mean equatorial plane at Julian year 2000.0, pointing towards the Northern Hemisphere
•X-axis: Vector pointing from the center of the Earth to the mean vernal equinox at Julian year 2000.0
•Y-axis: Vector perpendicular to the x- and z-axes, forming a right-handed coordinate system
•Origin: Center of the Earth

Notes:
•This is the principal coordinate system used for computations in FreeFlyer
•Commonly referred to as Mean of J2000, J2000, ECI, or GCI

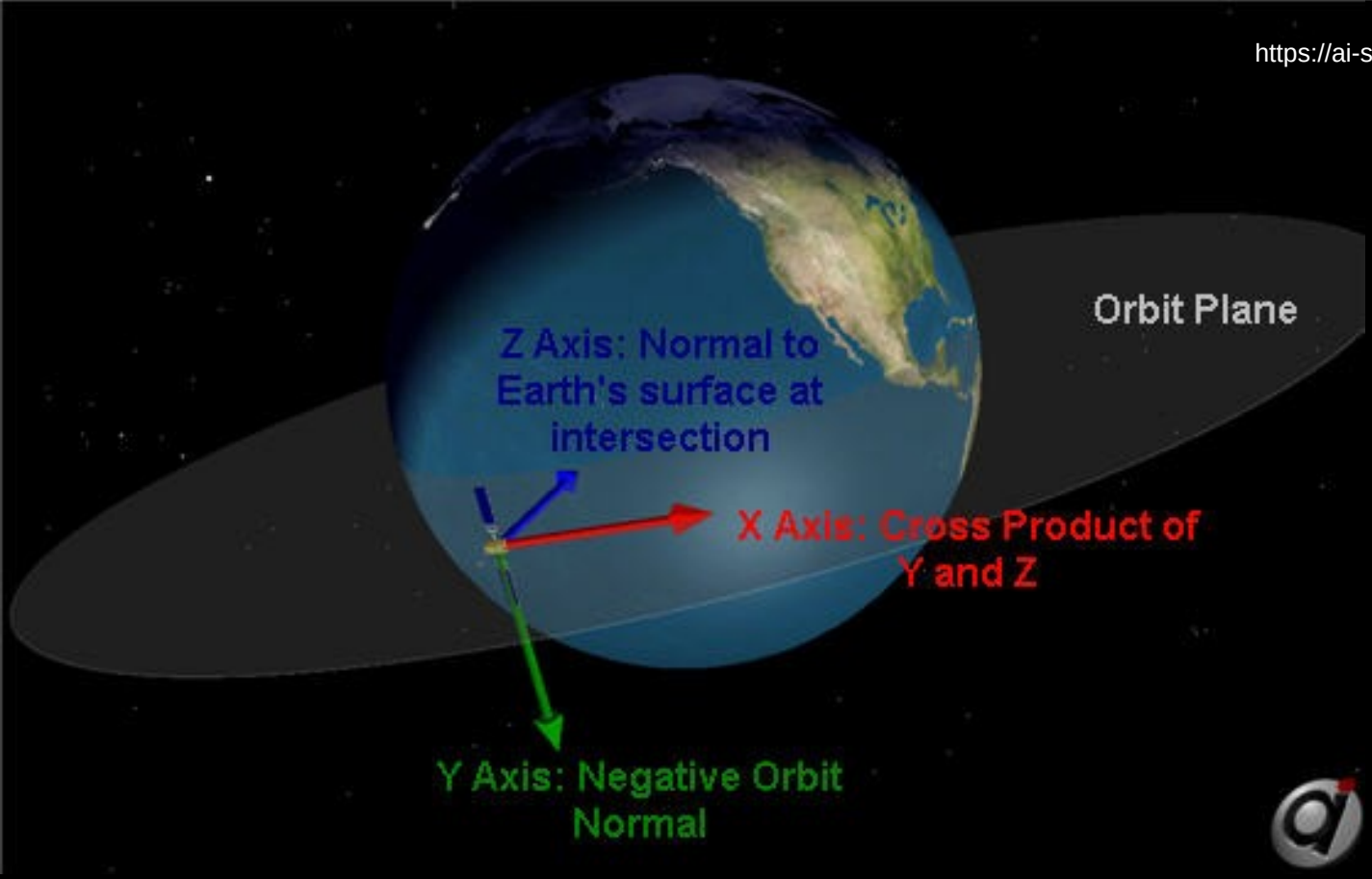
https://ai-solutions.com/_help_Files/attitude_reference_frames.htm



LVLH - Earth Pointing

Description:
•Z-axis: Vector pointing in the opposite direction to the position vector (points to center of Earth)
•Y-axis: Vector pointing in the opposite direction to the orbit normal (the orbit normal is the cross product of position and velocity)
•X-axis: Vector perpendicular to the y- and z-axes, forming a right-handed coordinate system
•Origin: Center of the Spacecraft

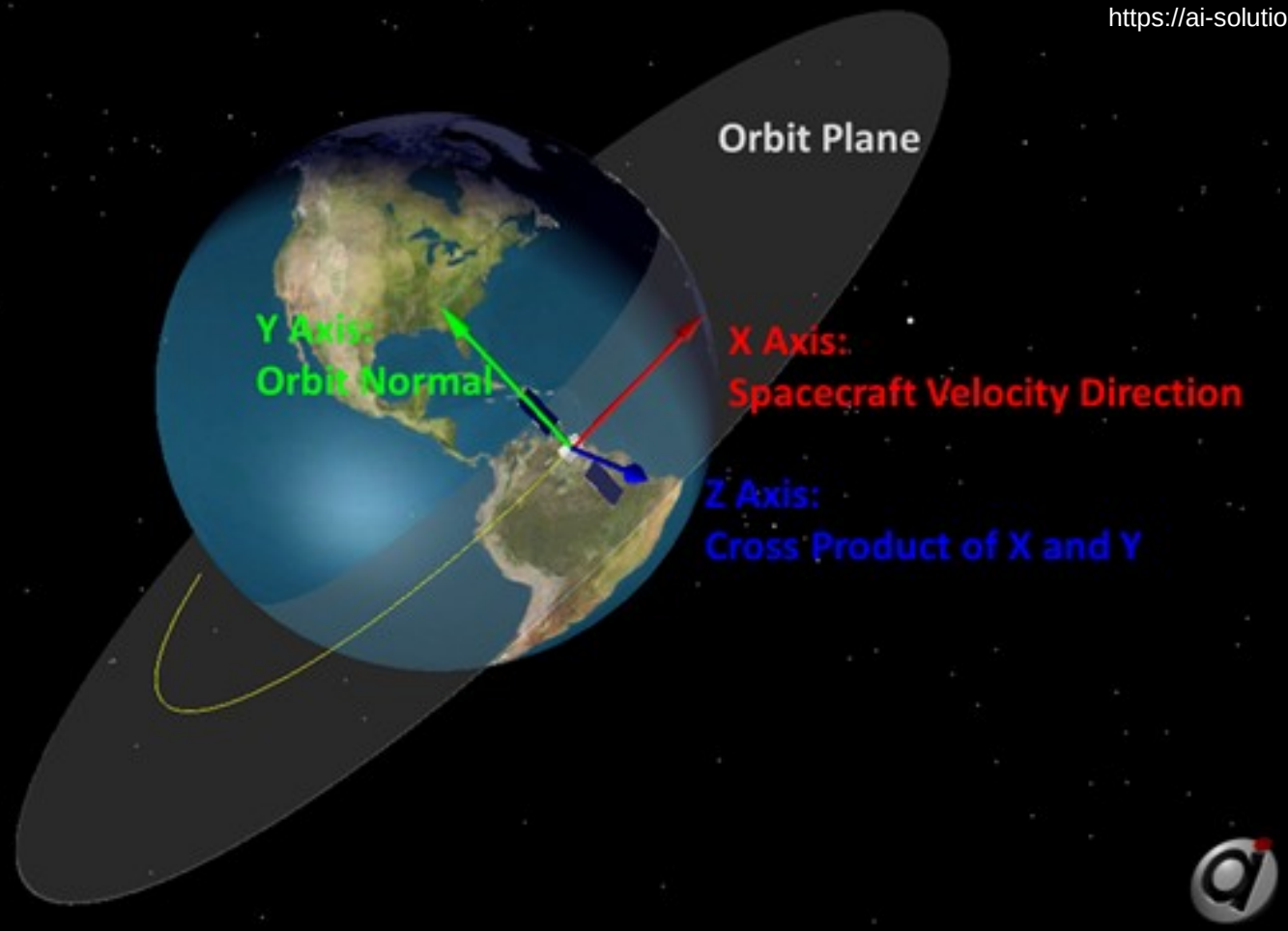
https://ai-solutions.com/_help_Files/attitude_reference_frames.htm



Geodetic

Description:
•Z-axis: Vector pointing toward Earth which is normal to the Earth's surface at intersection with Earth's surface
•Y-axis: Vector pointing the opposite direction of the orbit normal
•X-axis: Vector perpendicular to the y- and z- axes, forming a right-handed coordinate system
•Origin: Center of the Spacecraf

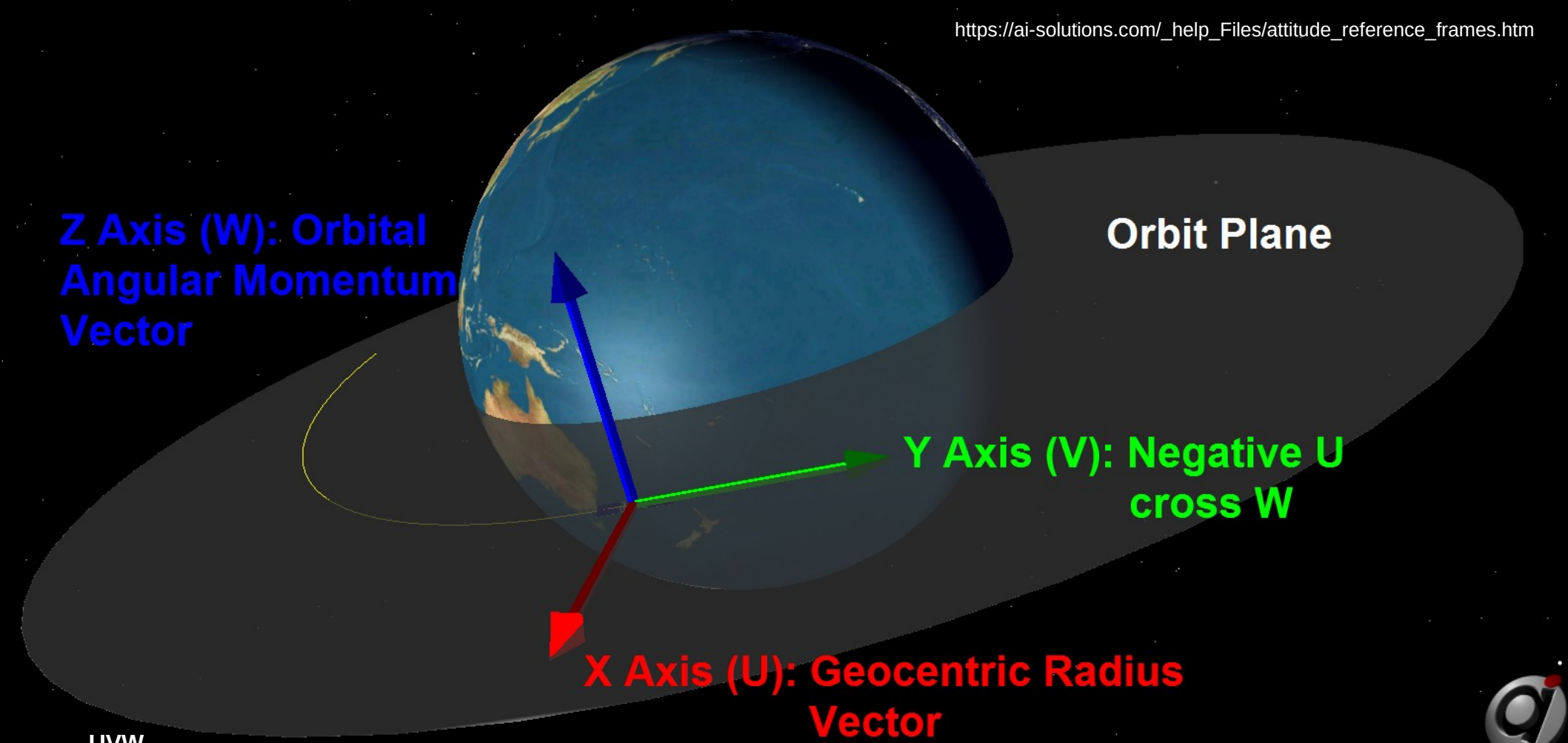
https://ai-solutions.com/_help_Files/attitude_reference_frames.htm



VNB

Description:
•X-axis: Vector pointing in the direction of the Spacecraft's velocity
•Y-axis: Vector pointing in the direction normal to the orbit
•Z-axis: Vector perpendicular to the x- and y-axes, forming a right-handed coordinate system
•Origin: Center of the Spacecraft

https://ai-solutions.com/_help_Files/attitude_reference_frames.htm

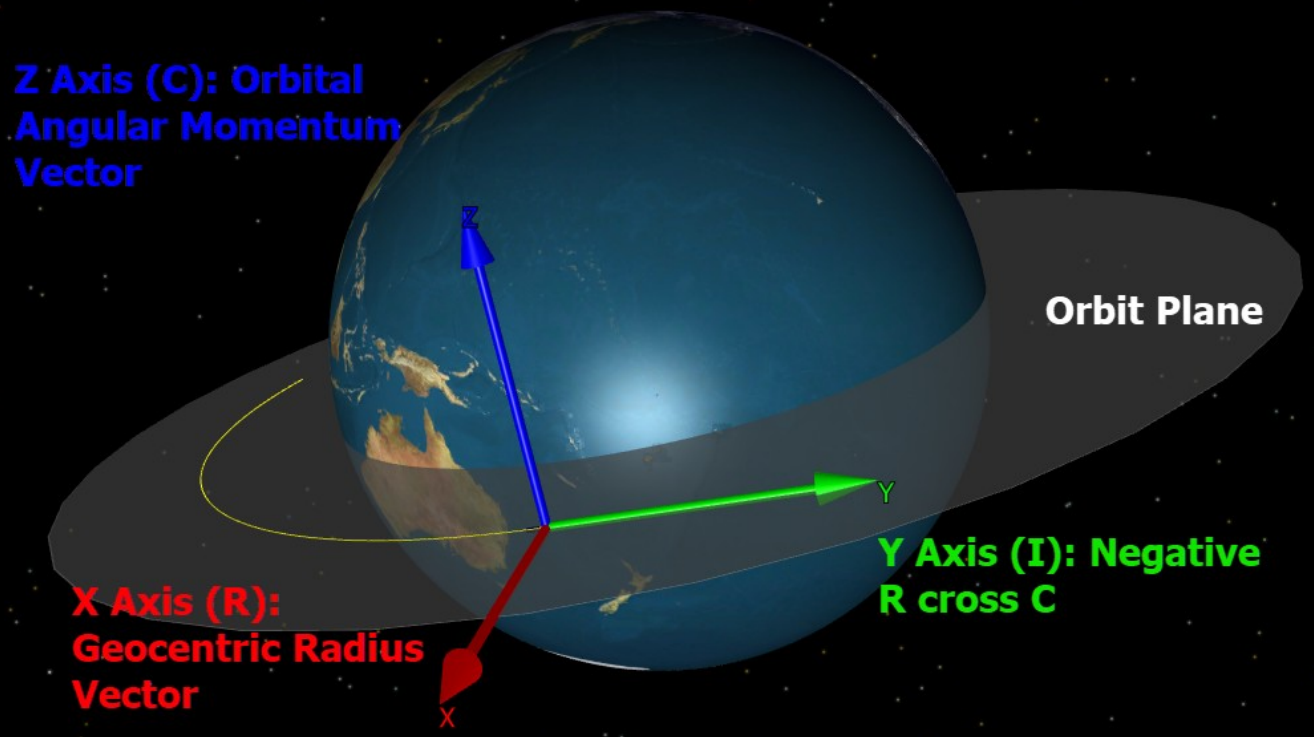


UVW

Description:
•Z-axis: Vector pointing in the direction normal to the orbit (along the angular momentum vector)
•X-axis: Vector pointing in the direction of the Spacecraft's position vector
•Y-axis: Vector perpendicular to the x- and z-axes, forming a right-handed coordinate system
•Origin: Center of the Spacecraft

Notes:
•The UVW coordinate system is quasi-inertial,
(for example, for the purpose of velocity conversions), but it is redefined at each point of interest.

https://ai-solutions.com/_help_Files/attitude_reference_frames.htm



RIC

Description:
•Z-axis: Vector pointing in the direction normal to the orbit plane (along the angular momentum vector, also commonly called "cross-track")
•X-axis: Vector pointing in the direction of the Spacecraft's position vector ("radial")
•Y-axis: Vector perpendicular to the x- and z-axes, forming a right-handed coordinate system (also commonly called "in-track")
•Origin: Center of the Spacecraft

Notes:
•The RIC coordinate system is aligned with the UVW coordinate system,
but is not considered quasi-inertial (it is not treated as an inertial coordinate system for the purposes of velocity vector conversions)

Ref.-Koor.-Sys. : LVLH

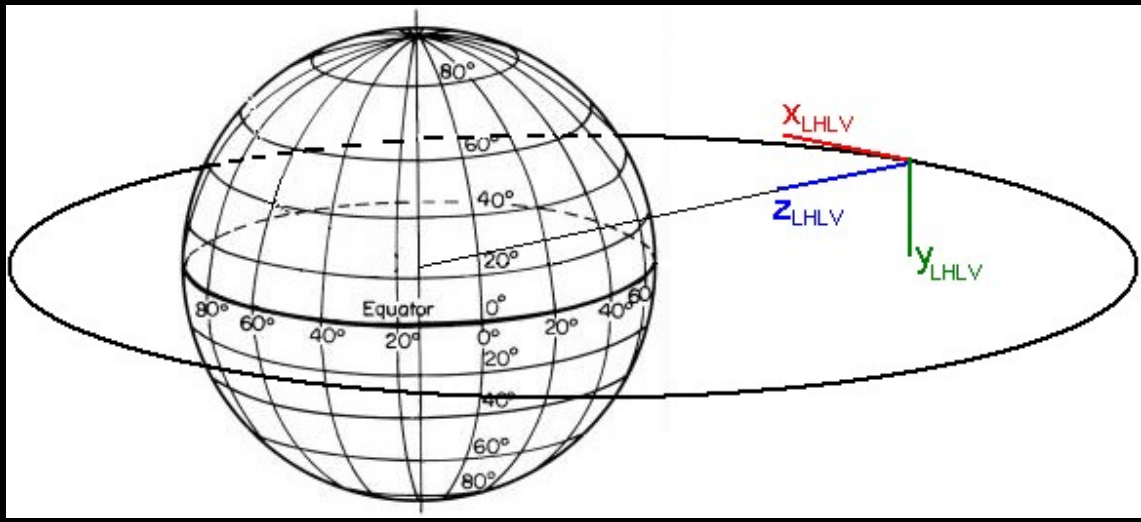
Local Vertical Local Horizontal (LVLH)
The origin of this frame is within the satellite.
The LHLV z-axis points nadir to the centre of Earth.
The X-axis is perpendicular to z and points in flight direction.
(It is a non inertial system, used for earth pointing)

Origin: Satellite Center of Mass

$+Z_{LVLH}$ Nadir to centre of Earth

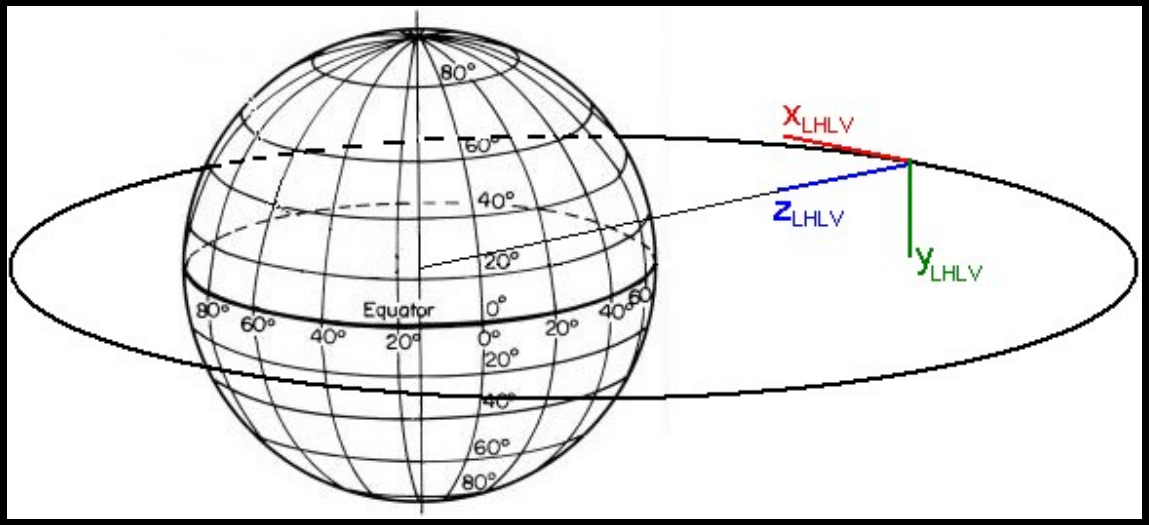
$+X_{LVLH}$ Perpendicular to Z_{LVLH} and in flight direction

$+Y_{LVLH}$ Building a right handed orthogonal system



Inertial, Stars, Sun & Earth

e.g.: Z: Directed to the sun,
X: Direction of the earth's movement on its orbit,
Y: Orthogonal (North Pole/South Pole? +/- 23°)



Celestial Coordinate System

For mapping positions on the celestial sphere. Using a system of spherical coordinates projected on the celestial sphere.

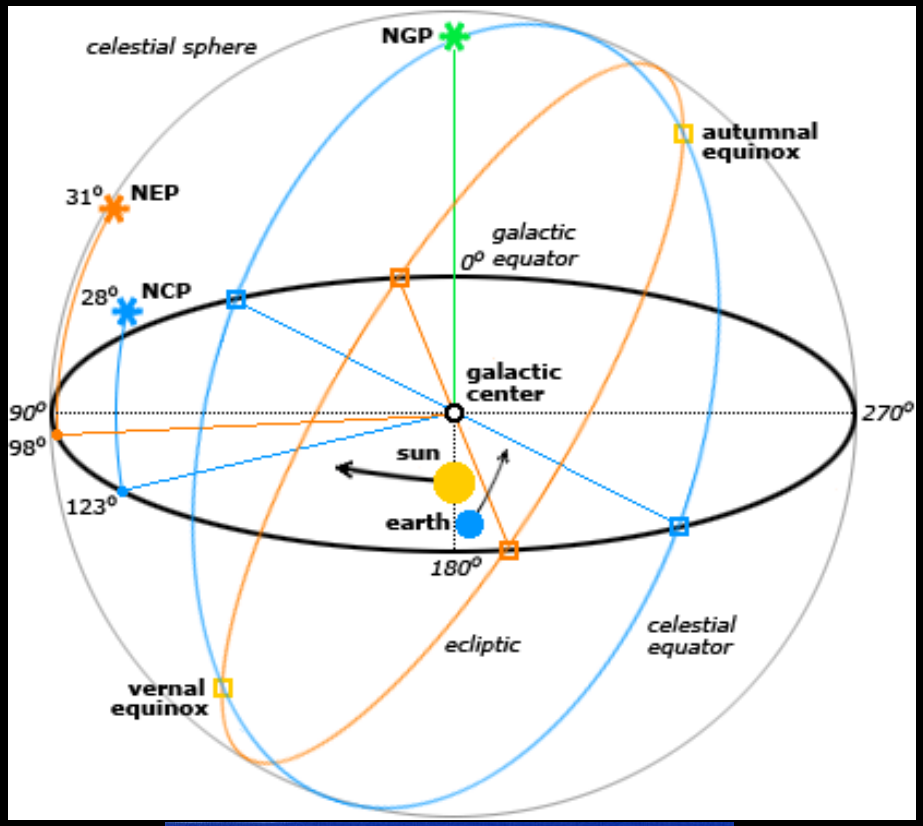
References:

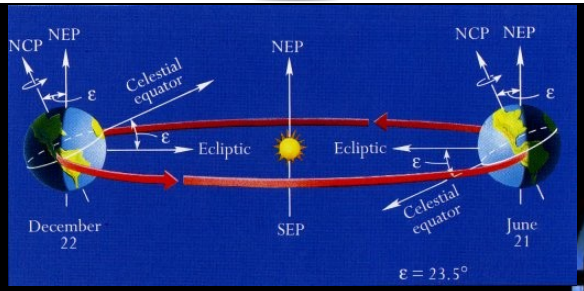
Horizontal system, based on the position of the observer on earth

Equatorial coordinate system, centered at Earth's center and fixes the sky

Galactic system, our solar system as the center, pointing towards the center of our galaxy

Supergalactic, its equator is aligned with the supergalactic plane, a major structure in the local universe, formed by nearby galaxy clusters

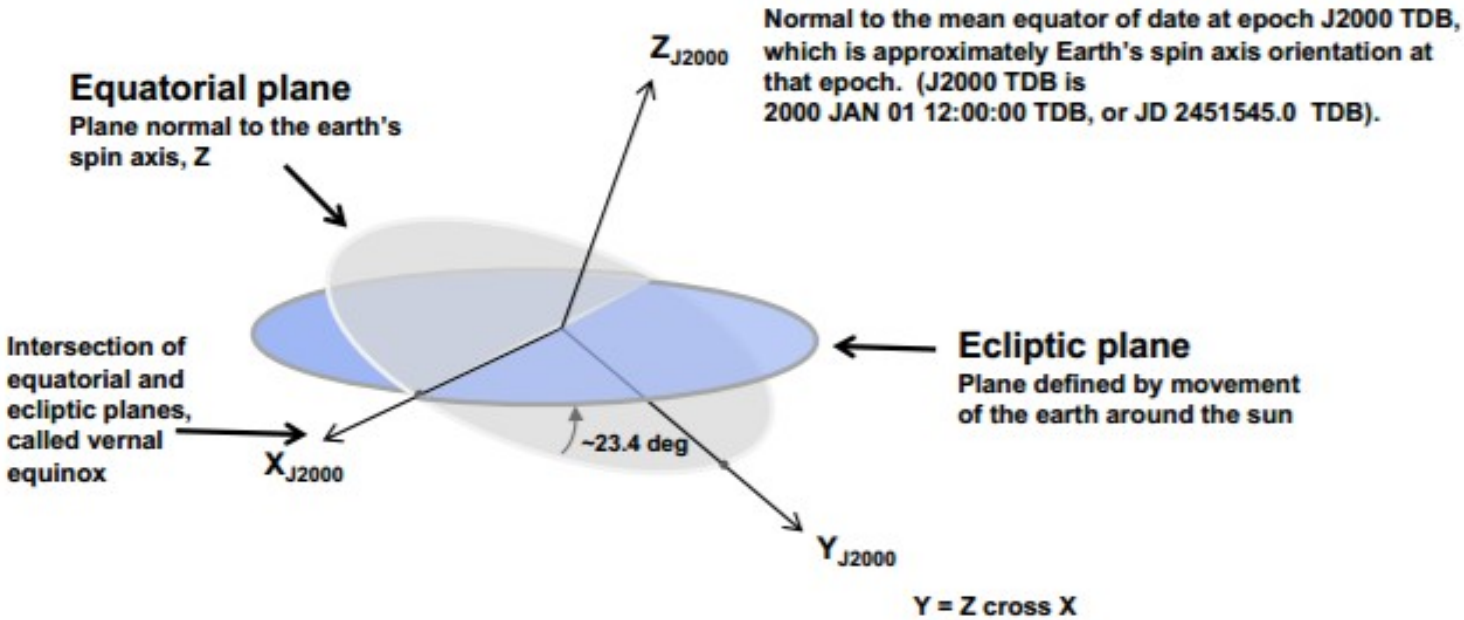




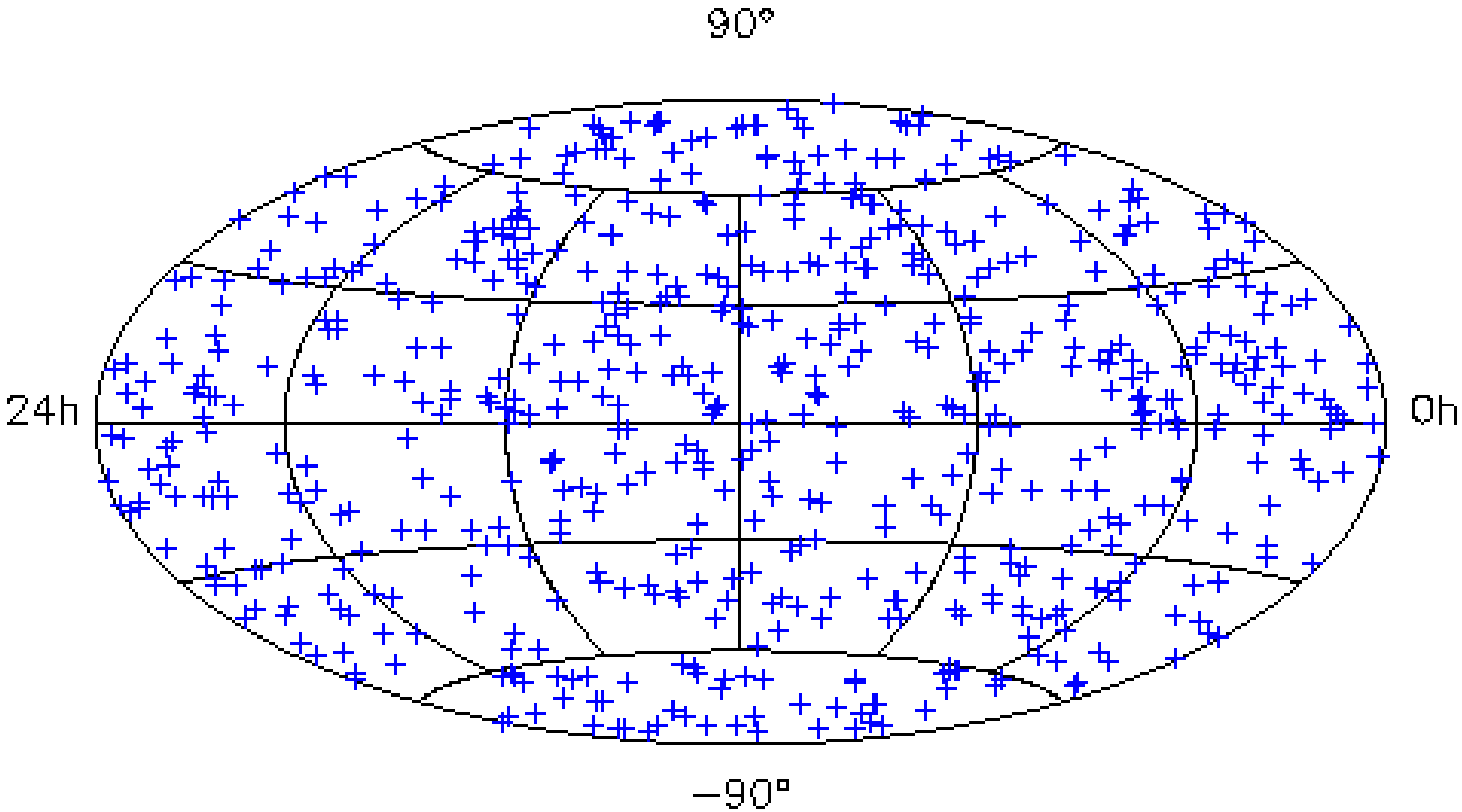
Stars Catalogue?

J2000 Frame

The J2000* frame definition is based on the earth's equator and equinox, determined from observations of planetary motions, plus other data.

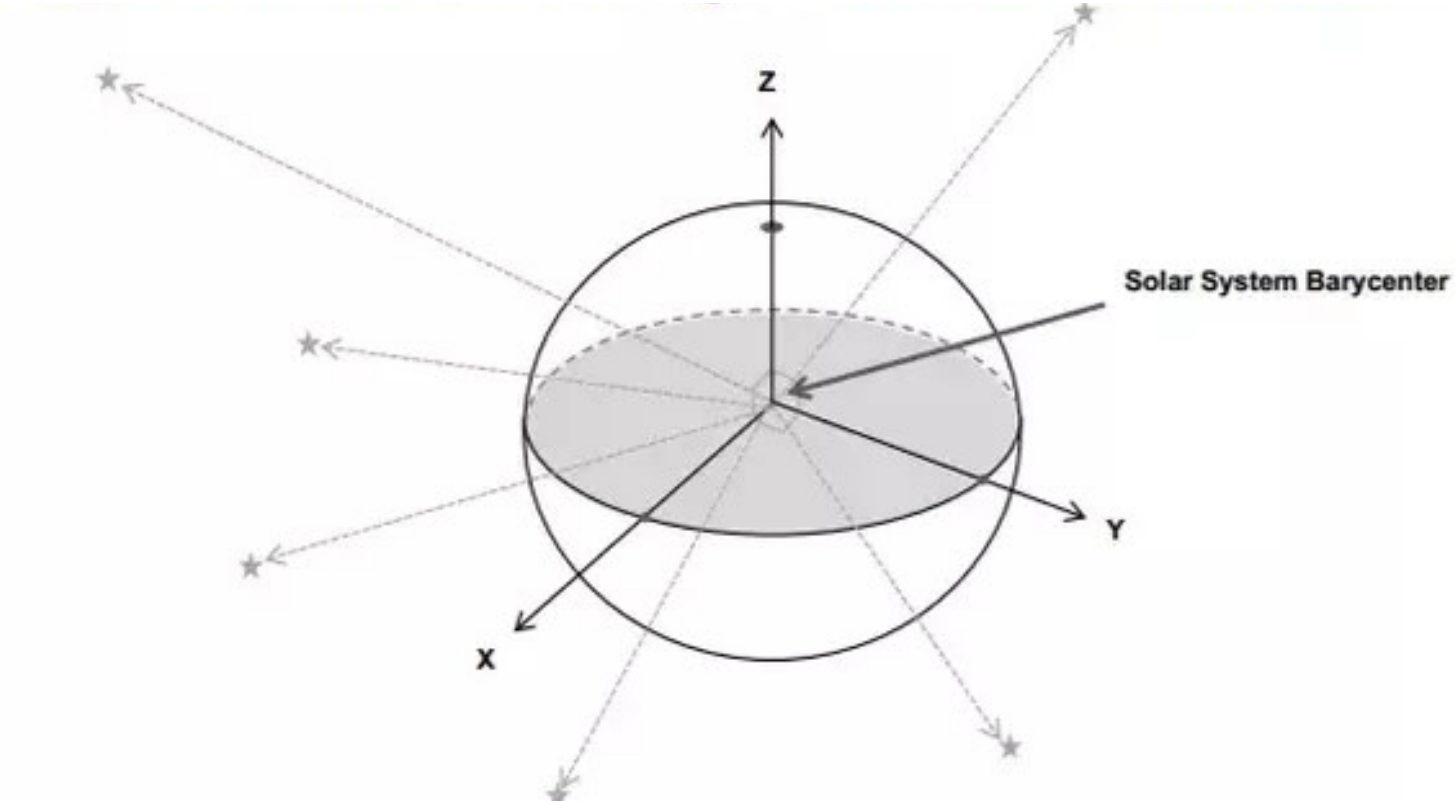


Stars Catalogue



„Real-Time“ Extragalactic Ref.-System

ca. 300 extragalactic Radio-Sources

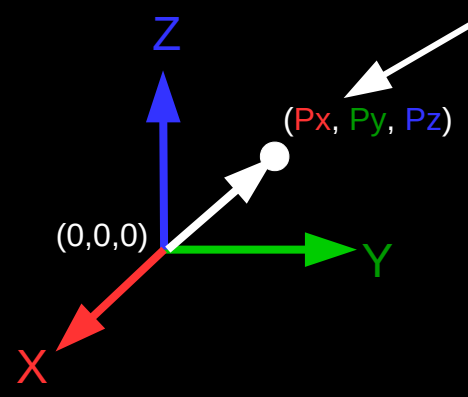


Back to

- Position
- Attitude
- Translation
- Rotation
- Pose
- Δ Pose

Position: Vector in which Coordinate-System?

Cartesian System



One point! Which?

Body-System: anchored in structure

Reference-System: **independent** of own rotation

(That's how it should be... but...)

For Position: One Point

For landing: front landing gear

When calculating with rotations, then use Center of Gravity (COG)

Where is the Plane? cm-precisely!

GPS antenna

(x,y,z) distance of GPS antenna to the wheels sufficient?
Translation + Rotation (relative to what?: Attitude)
Center of Gravity? Changes!

Distance to ground?

GPS: „Yet 30m to ground“

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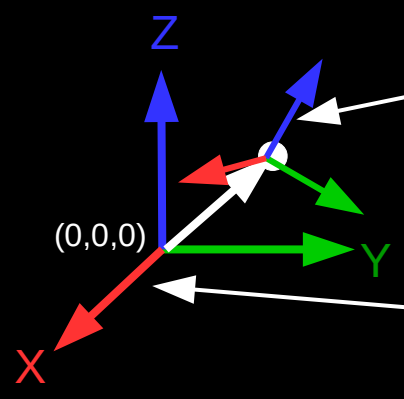
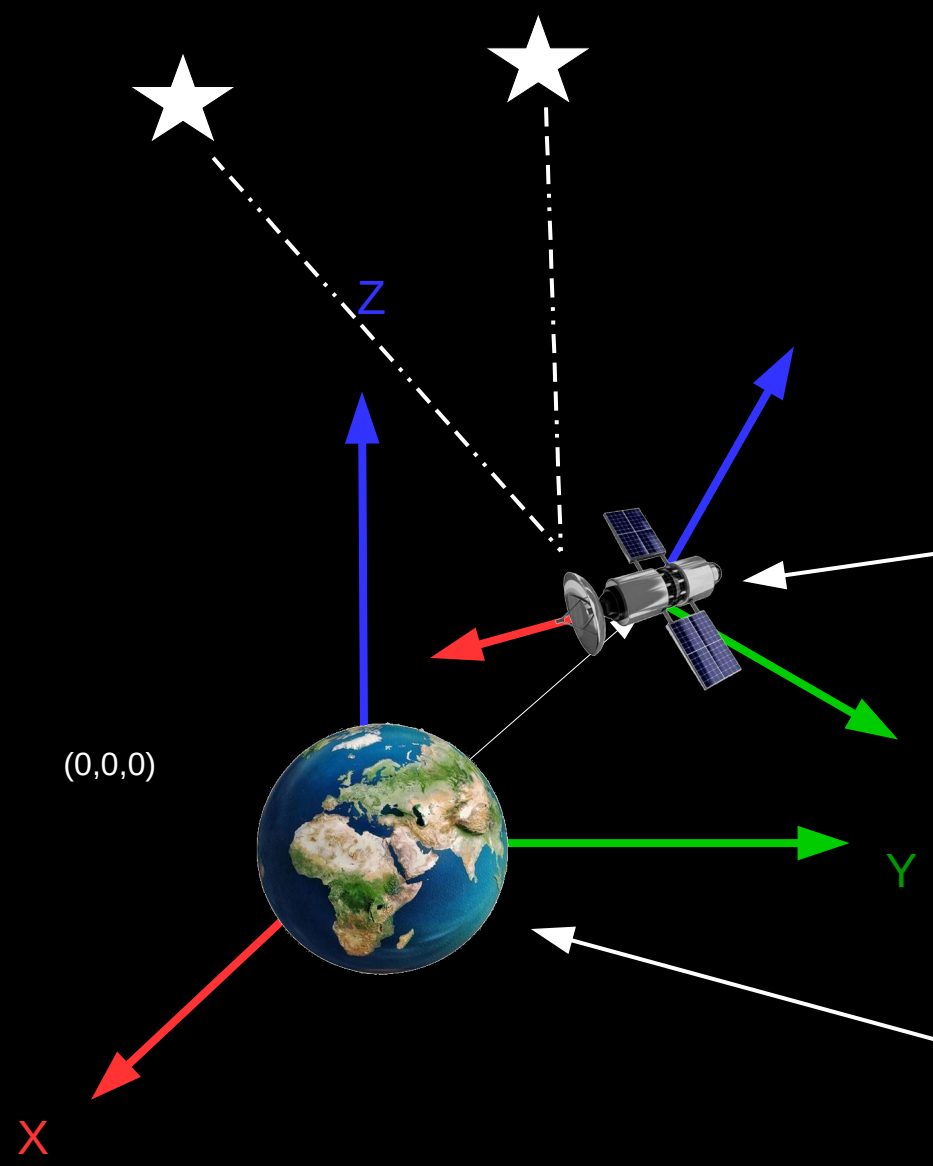
Attitude: Angle between what?

Angular deviation of two coordinate systems
A point doesn't have an attitude!
A vector has an 2D-Attitude – Which value is missing?



Body-System: anchored in structure
For 3D-Attitude:
we need at least 2 vectors

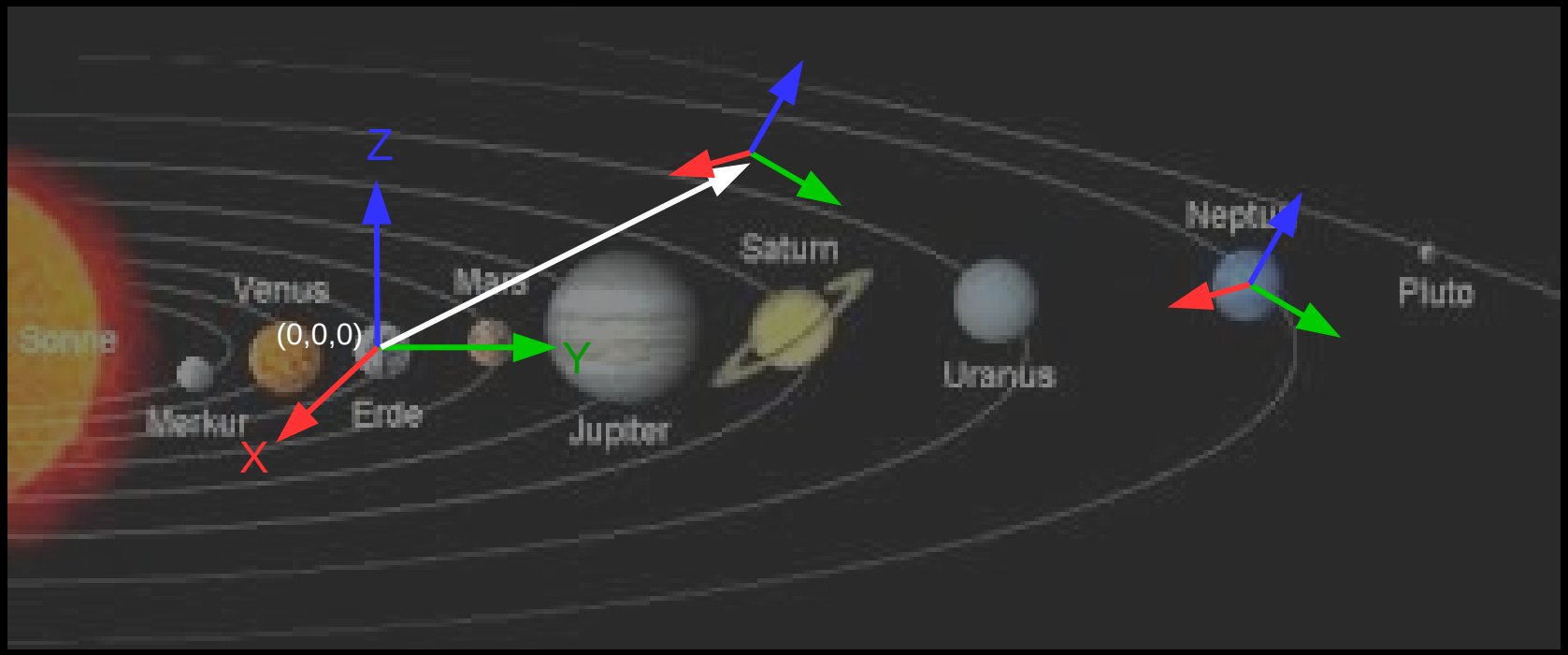
Reference-System:
independent of own rotation



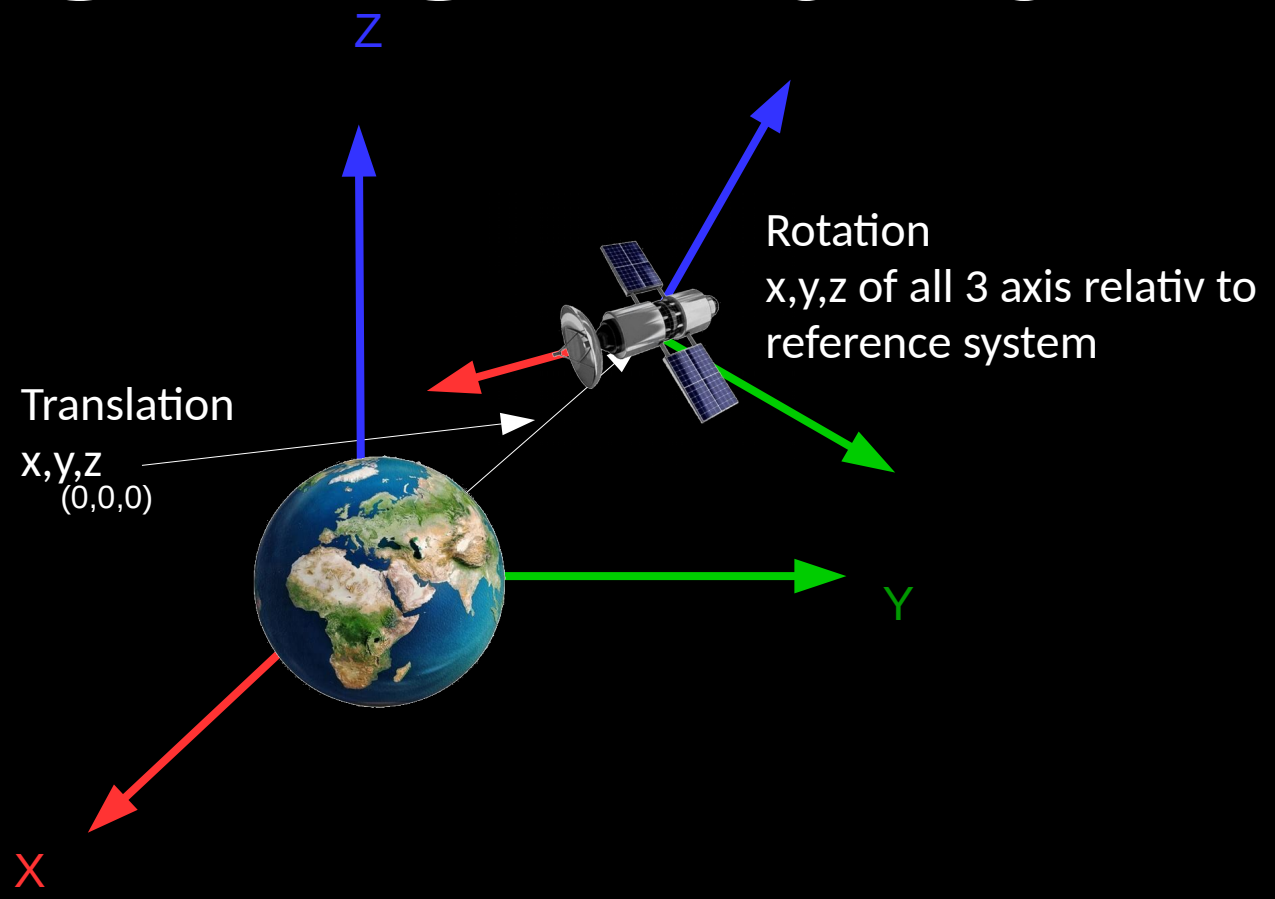
Body-System: anchored in structure
For 3D-Attitude:
we need at least 2 vectors

Reference-System:
independent of own rotation

Attitude

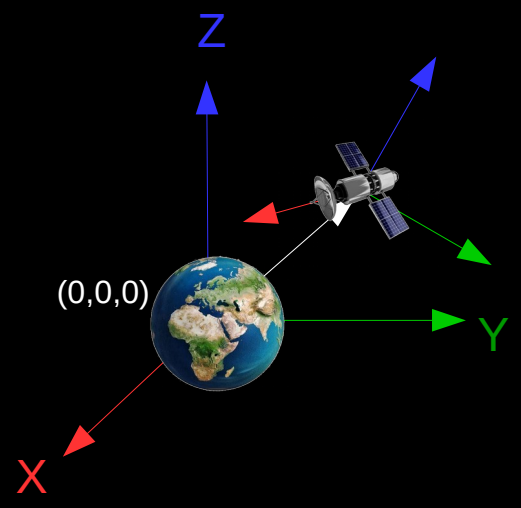

Z
Y
X
(0,0,0)
Sonne
Merkur
Venus
Erde
Mars
Jupiter
Saturn
Uranus
Neptun
Pluto

Pose: Position + Attitude
3D + 3D = 6 DoF



Notice!

All pose specifications und transformations (Δ Pose) are the relation between two coordinate systems



For position und translation:

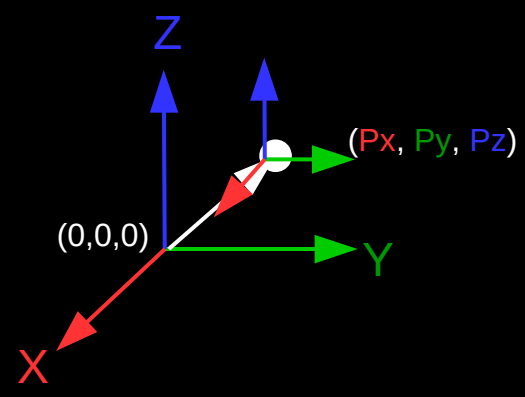
Reference-Coord.-Sys. and one point,
e.g. (0,0,0) of the Body-CS

For attitude und rotation:

Axis of Reference-Coord.-Sys. and
Axis of Body-CS
At least 2 vectors (with only one vector: 1D remains undefined)

Translation

Translation: x = x1+x2; y = y1+y2; z = z1+z2



Body-Coordinates

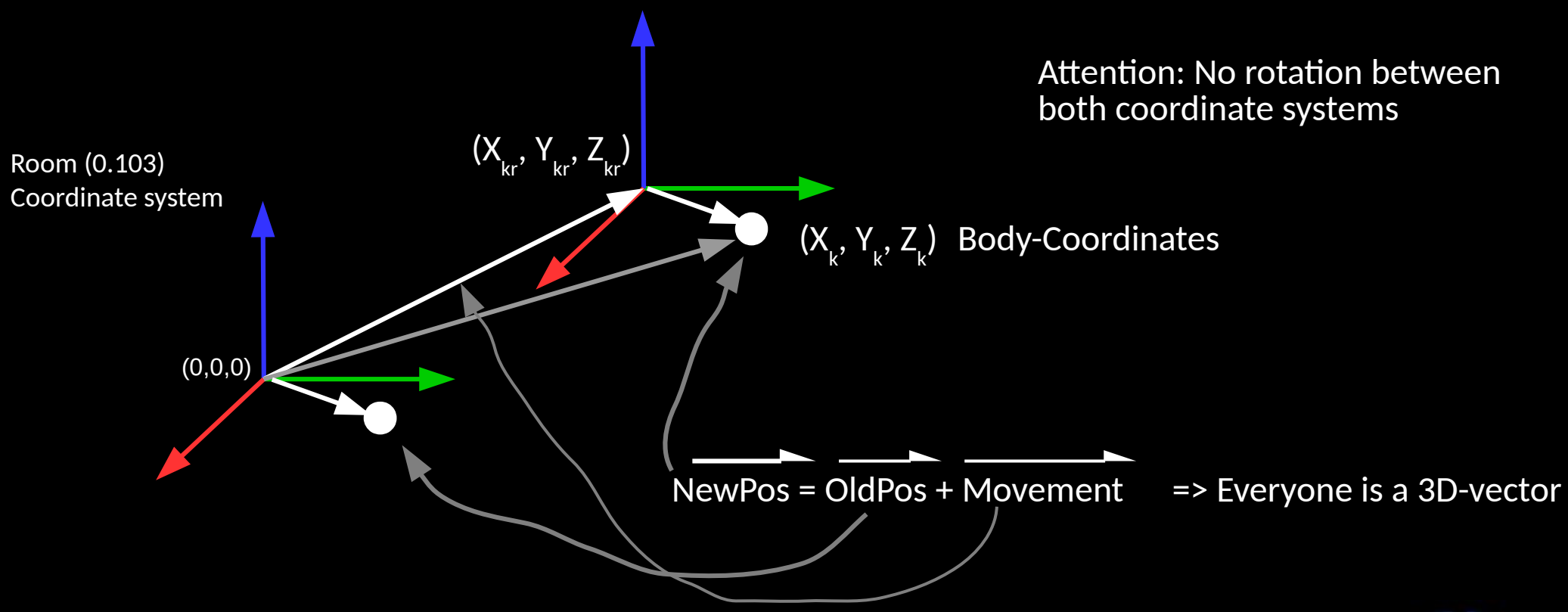
(X_k, Y_k, Z_k) e.g. button “A” on my laptop

$\overrightarrow{\text{NewPos}} = \overrightarrow{\text{OldPos}} + \overrightarrow{\text{Movement}}$ => each one is a 3D-vector

Translation

Translation: x = x1+x2; y = y1+y2; z = z1+z2

Attention: No rotation between both coordinate systems

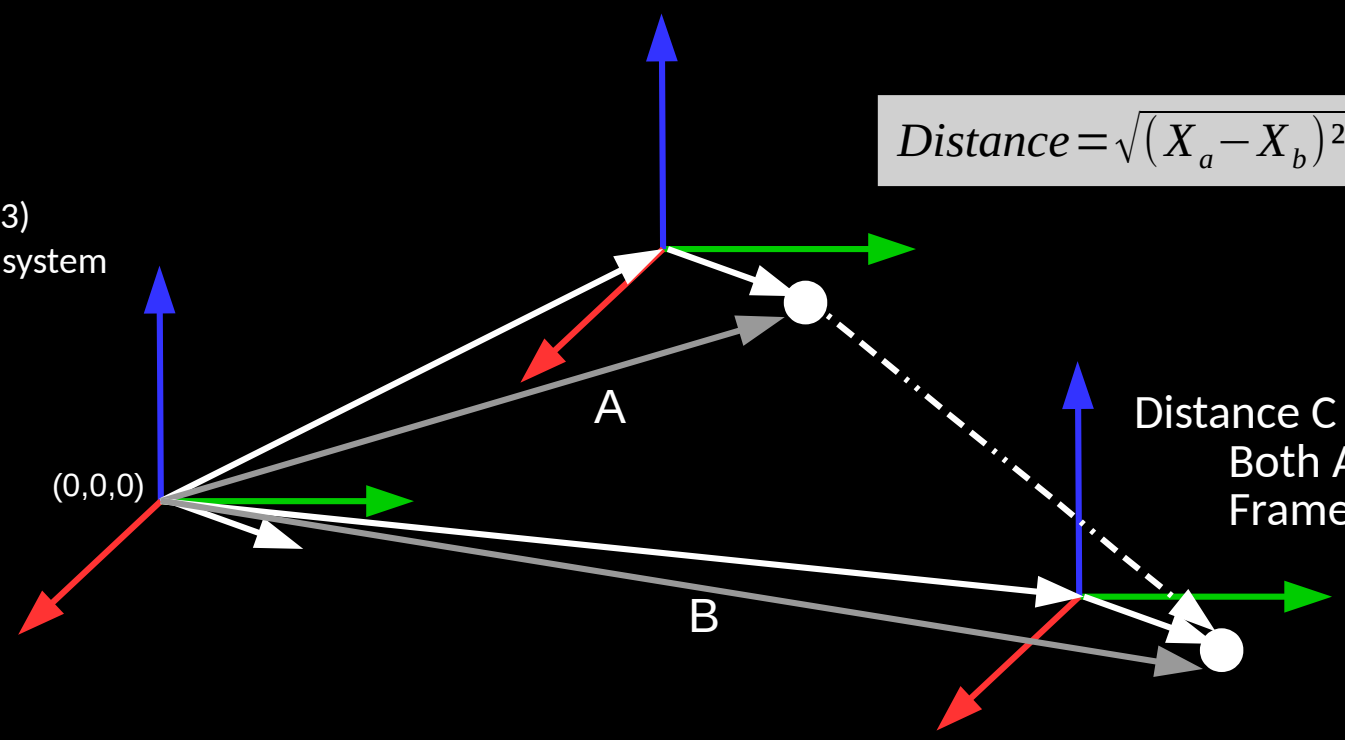


Translation

Translation: x = x1+x2; y = x1+y2; z = z1+z2

$$Distance = \sqrt{(X_a - X_b)^2 + (Y_a - Y_b)^2 + (Z_a - Z_b)^2}$$

Room (0.103)
Coordinate system



Distance C = | A - B |
Both A and B relativ to same Frame